Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation

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Abstract

We develop an estimation theory on finite-grid path space for stochastic systems whose two-time transition families violate the Chapman–Kolmogorov factorization and therefore admit no generator on the observed state space whose evolution reproduces the given family. Candidate paths supplied by an external proposal mechanism are reweighted by the exponential of a discretized Onsager–Machlup (OM) action computed against a tractable surrogate model that is explicitly not the truth, and expectations under the resulting Gibbs-selected measure are estimated by self-normalized importance sampling (SNIS). Up to a constant and the OM divergence term, the selection weight is the surrogate's own discrete path density raised to the temperature, so the selected measure is a *product* of two path densities rather than the density *ratio* used by Girsanov reweighting; this identity is the construction's point of departure from the path-reweighting literature. For Gaussian pairs the product adds precisions to leading order in the grid step δt: the selected measure acquires the diffusion coefficient $1/\sigma_{\mathrm{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$, different from the proposal's for every temperature $\lambda > 0$, and this quadratic-variation mismatch is the mechanism behind every degeneracy the paper reports—the chi-square divergence between selected measure and proposal grows exponentially under grid refinement, the selected measures converge to a continuum law that is singular to the proposal's, and the finite grid is the primary setting because the importance-sampling representation of the selected measure from the proposal does not survive refinement. The estimator theory is standard SNIS theory specialized to this weight and is stated as such: strong consistency, a central limit theorem, finite-sample concentration in two tiers (polynomial under finite chi-square divergence, exponential under bounded weights), and degeneracy diagnostics organized by the chi-square divergence. The relation to the true process is a conditional transfer under an explicit total-variation hypothesis, flanked by an unconditional kernel-level obstruction bounding, under a contraction hypothesis, every divisible surrogate family away from an indivisible target; the two are logically independent, and under grid refinement the hypothesis is attainable only for proposal–surrogate–temperature triples designed to the target's quadratic variation. Surrogate calibration is formulated as Euler quasi-likelihood estimation, and the inconsistency of calibrating by the OM action itself is recorded. Two fully worked one-dimensional examples are given: a Gaussian pair in which every quantity is in closed form, and a double-well surrogate with non-constant drift divergence in which the benchmark is obtained by transfer-matrix quadrature and the divergence term measurably changes the estimand.

Keywords. Onsager–Machlup functional, selection functional, self-normalized importance sampling, Gibbs reweighting, indivisible stochastic processes, non-Markovian stochastic processes,

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Significance statement

This paper provides estimation methodology for expectations under a Gibbs-selected path measure built from two ingredients: an externally supplied ensemble of candidate paths on a finite time grid, and a discretized Onsager–Machlup action computed against a deliberately imperfect surrogate model. The motivating setting is a stochastic system whose two-time transition operators violate the Chapman–Kolmogorov relation, so that no Markov generator on the observed state space reproduces them and the generator-based path-density derivations of classical stochastic modeling are unavailable on that description. The target family does not enter the estimator's construction or its guarantees; it enters only the conditional transfer of Theorem D. The self-normalized importance-sampling estimator consistently evaluates expectations under the selected measure, with central-limit, finite-sample concentration, and effective-sample-size guarantees. The selected measure is identified as a product of the proposal law and the tempered surrogate path law; this fixes its diffusion coefficient and explains why the construction is intrinsically a finite-grid object. The connection to the true process is a conditional transfer under an explicit hypothesis, together with an unconditional impossibility result bounding every divisible surrogate away from an indivisible target's kernels. The paper does not claim uncertainty quantification for the true non-Markovian system; it quantifies uncertainty under the selected measure, and it states the gap between the two as an open problem in a form that specifies what would have to be controlled.

1 Introduction

The Onsager–Machlup (OM) action functional has two classical operational roles, both predicated on the existence of a generator. For an Itô stochastic differential equation (SDE)

$$dX_t \;=\; b(X_t)\,dt \;+\; \sigma\,dW_t, \qquad t\in[0,T],\; X_t\in\mathbb{R}^d, \tag{1}$$

with non-degenerate diffusion $\sigma\sigma^\top \succ 0$, the OM action of an absolutely continuous trajectory $\xi\in C([0,T];\mathbb{R}^d)$,

$$S^{\mathrm{OM}}[\xi] \;=\; \frac{1}{2}\int_0^T \left|\sigma^{-1}(\dot\xi - b(\xi))\right|^2 dt + \frac{1}{2}\int_0^T \nabla\cdot b(\xi)\,dt, \tag{2}$$

serves either (i) as the asymptotic relative density of the law of X on ε-tubes around ξ in path space (Dürr–Bach [25]; Ikeda–Watanabe [44]; Capitaine [12]; Pinski–Stuart [64]; Pinski–Stuart–Theil [65]; Selk–Honnappa [67]; Ayanbayev–Klebanov–Lie–Sullivan [4]), or (ii) as the Freidlin–Wentzell rate functional governing small-noise large deviations [29, 20, 34]. Recent extensions cover stochastic partial differential equations with Lévy noise [41], jump-diffusions [14, 42], and time-varying noise [78]. In every case the action takes the drift–diffusion pair (b,σ) as an input.

For *indivisible* stochastic processes [5, 6, 55, 69, 70, 71] this input is unavailable. An indivisible target on a configuration space Ω is specified by a two-time column-stochastic kernel family $\{\Gamma(t\leftarrow s)\}_{0\le s\le t\le T}$ that violates the Chapman–Kolmogorov (CK) factorization, in that

$$\Gamma(t\leftarrow s) \;\neq\; \Gamma(t\leftarrow r)\,\Gamma(r\leftarrow s) \qquad \text{for some } s\le r\le t. \tag{3}$$

By Proposition 3.6 of Summers (Paper A) [70], no strongly measurable, Bochner-integrable family of bounded generators reproduces such a family through a time-ordered exponential on $L^2(\Omega, \mu)$; and since an Itô diffusion is Markov while Markov transition kernels satisfy Chapman–Kolmogorov, no drift–diffusion pair (b, σ) on Ω produces (1) with these two-time kernels either (Proposition 2.1). The exclusion covers exactly what the proof uses: CK failure of the observed family excludes a generator *on the observed state space* reproducing that family. It excludes neither a Markovian enlargement with hidden variables nor a path-space law (Remark 2.2); earlier versions of this paper overstated it.

The construction The discretized form of (2), computed against a chosen surrogate pair (b_ν, σ_ν) that is explicitly *not* the true dynamics, is used as a *selection functional* on a separately supplied ensemble of candidate paths on a finite time grid: each candidate path ξ receives the weight $w_\nu(\xi) = \exp(-\lambda S_{\nu,M}[\xi])$, and the weighted ensemble average of a path observable is a self-normalized importance-sampling (SNIS) estimator of its expectation under the Gibbs-selected measure $dQ_{\nu,M} \propto w_\nu \, d\Pi_M$. The proposal mechanism Π_M is abstract: any procedure generating grid paths with i.i.d. sampling is a valid input. The central claim, which Theorems A–C prove, is that *the estimator consistently evaluates observables under the path measure selected jointly by the proposal and the surrogate action.* That measure is not, and is nowhere claimed to be, the law of the target process; the target family enters neither the construction nor Theorems A–C, and its relation to the selected measure is treated separately in §7.

The weight Reweighting an externally generated path ensemble by a path action is established practice: Donati, Hartmann and Keller [24] reweight ensembles generated under a reference potential to averages under a perturbed one through the Girsanov change of measure, and Adib [1] does the same with discretized stochastic actions. Those constructions form the *ratio* of two path densities—the difference of two actions—and their weights are likelihood ratios. The present construction uses a *single* action. By Proposition 3.5, for constant-divergence surrogates $e^{-\lambda S_{\nu,M}}$ is, up to a constant, the surrogate's own Euler path density raised to the power λ, so the selected measure is a *product* of two path densities, $Q_{\nu,M} \propto \Pi_M \cdot p_\nu^\lambda$, a product of experts [39] on path space. Multiplying Gaussian path densities adds precisions to leading order in the grid step, and the selected measure acquires the diffusion coefficient $1/\sigma_{\text{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$ (Corollary 3.6). This is the construction's point of departure from the reweighting literature and the mechanism behind everything distinctive in its behavior: selected measure and proposal have different quadratic variation, so they become mutually singular under grid refinement; the chi-square divergence between them grows exponentially in the grid size; the selected measures converge to a continuum law, explicit in the Gaussian case, that no importance-sampling representation from the proposal can reach; and the truth-link hypothesis is attainable under refinement only for proposal–surrogate–temperature triples designed to the target's quadratic variation (§7). These properties are intrinsic to the construction and are not treated as defects.

Why the finite grid is the primary setting Two facts force it. The pointwise continuous-path weight is degenerate: under diffusion proposals, sample paths are almost surely not absolutely continuous, so $w_\nu = 0$ almost surely and the estimator is $0/0$ (Proposition 2.3). The quadratic-variation mismatch of Corollary 3.6 makes selected measure and proposal mutually singular in the limit: the selected measures converge, but to a law that no importance-sampling representation from the proposal can reach, so the fixed-proposal estimator has no nondegenerate limit under refinement (§9.4). All statements are therefore made on a fixed finite grid, where the action is a

finite sum, the weights are positive, and the selected measure exists unconditionally. The grid is the model, not a truncation of one; grid size, temperature, proposal and discretization convention are all part of the definition of the estimand (§9.6). The intended applications are settings with an external candidate-path generator on a state description whose empirical two-time family fails Chapman–Kolmogorov—turbulent boundary-layer transitions with documented Mori–Zwanzig memory [51, 52, 32, 68], and metastable molecular dynamics without time-scale separation [36, 60]. The target family has two operational roles: the proposal simulator is built to approximate it, and the surrogate is calibrated from data it produces (§8.4). It plays no role in the estimator's guarantees.

Contributions We separate what is standard from what is the paper's own.

Standard, and used as such. The SNIS estimator, its strong consistency, central limit theorem and asymptotic variance (Theorem A); Chebyshev–Cantelli and Bernstein–Hoeffding concentration (Theorem B); the chi-square characterization of the effective sample size and the Rényi ordering $\exp(D_{\mathrm{KL}}) \le 1 + \chi^2$ (Theorem C); the Gibbs form of the selected measure; the Chatterjee–Diaconis and Sanz-Alonso sample-size scales. Each is cited at its point of use and is specialized to the selection weight, not extended.

The paper's own.

(C1) The product identity for the selection weight (Proposition 3.5) and its Gaussian corollary (Corollary 3.6): the selected measure is a product of the proposal law and the tempered surrogate path law, with diffusion coefficient $1/\sigma^2_{\mathrm{eff}} = 1/\sigma^2_{\Pi} + \lambda/\sigma^2_{\nu} + O(\delta t)$; hence λ is a diffusion-coefficient control, not only a temperature, and for a Gaussian Markov proposal the selected measure is itself the law of a Markov chain.

(C2) The finite-grid formulation adopted because it is forced on the estimator: by the pointwise degeneracy of Proposition 2.3 and by the mutual singularity of selected measure and proposal under refinement, quantified in §9.4 and contrasted there with the bounded chi-square of Girsanov ratio weights on the same grids.

(C3) The two-tier concentration structure matched to the two weight regimes that occur, together with the observation that the exponential tier is vacuous in Example I and inapplicable in Example II (Remark 5.4).

(C4) The truth link stated as a conditional transfer (Theorem D) paired with an unconditional kernel-level obstruction (Proposition 7.2), with their logical independence made explicit by example (Example 7.4) and the attainability of the transfer hypothesis under refinement constrained by a design rule on the quadratic variation of the proposal–surrogate–temperature triple, proved for Gaussian pairs (§7.5).

(C5) Calibration formulated as Euler quasi-likelihood M-estimation (§8.4), with the inconsistency of calibrating by the OM action itself recorded: in the worked example the action-minimizer returns a restoring rate of 2.76 for a true rate of 1.

(C6) Two fully worked examples (§9): a Gaussian pair in which the estimand, the chi-square and Kullback–Leibler divergences, and the asymptotic variance are in closed form, with a 1,000-replication convergence study; and a double-well surrogate with non-constant divergence, benchmarked by transfer-matrix quadrature, in which the OM divergence term moves the estimand by 0.09 and the finite-chi-square condition is certified by a one-line criterion (Lemma 8.1).

Relationship to the program This paper is the estimation-methodology member of a program on indivisible stochastic dynamics [69, 70, 71]. From Paper A [70] it imports four self-contained items, cited at their point of use: the deficit operator $\Delta(t,s,r)$ and its norm $D(t,s,r)$, the no-generator proposition (Proposition 3.6 there), the three-pair obstruction bound under a standing contraction assumption (Corollary 4.6 there), and the integrated deficit $J(\mathcal{R})$ (Theorem 7.4 there). Whether invariants of the target control the truth-link tolerance is the open problem of §7.5, stated there in design form. No dilation-theoretic machinery is used.

Outline §2 recalls the classical OM action, states the no-input proposition with its corrected scope, and proves the pointwise degeneracy. §3 formalizes the grid construction and proves the product identity. §4–§6 state Theorems A–C. §7 states Theorem D, the obstruction, their independence, and the design rule. §8 characterizes three surrogate families and formulates calibration. §9 works two examples against exact benchmarks and states the grid dependence of the estimand. §10 positions the construction against path reweighting, misspecified-model theory, variational inference, and generalized Bayes. §11 states what is open.

2 The classical Onsager–Machlup action and its two breakdowns

We work on a finite horizon $[0,T]$. Let $X = (X_t)_{t\in[0,T]}$ be the Itô diffusion (1) on $\mathbb{R}^d$, with $b \in C_b^1(\mathbb{R}^d;\mathbb{R}^d)$, constant non-degenerate $\sigma \in \mathbb{R}^{d\times d}$, and initial value $X_0 = x_0$. The infinitesimal generator $L = b\cdot\nabla + \frac{1}{2}\sigma\sigma^\top : \nabla^2$ acts on a dense subspace of $C_b(\mathbb{R}^d)$. The law μ_X of X on $C([0,T];\mathbb{R}^d)$ is equivalent to the law of $x_0 + \sigma W$, with Radon–Nikodym density given by Girsanov's theorem.

Tube density Fix a reference path $\xi^\circ \in C([0,T];\mathbb{R}^d)$ that is absolutely continuous with $\xi^\circ(0) = x_0$ and $\dot\xi^\circ \in L^2$. Dürr–Bach [25] and Ikeda–Watanabe [44, Section VI.9] show that, under the regularity above and for absolutely continuous ξ with $\xi(0) = x_0$ and $\dot\xi \in L^2$, the sup-norm tube probabilities satisfy

$$\lim_{\varepsilon\downarrow 0} \frac{\mu_X(\{\eta : \sup_t |\eta_t - \xi_t| \le \varepsilon\})}{\mu_X(\{\eta : \sup_t |\eta_t - \xi^\circ_t| \le \varepsilon\})} \;=\; \exp\bigl(-(S^{\mathrm{OM}}[\xi] - S^{\mathrm{OM}}[\xi^\circ])\bigr). \tag{4}$$

The two ingredients of S^{OM} both encode the generator: b enters through the kinetic term $|\sigma^{-1}(\dot\xi - b)|^2$ and through the divergence term $\nabla\cdot b$. The divergence term is not displayed by the Girsanov density in its Itô form $\exp(\int \sigma^{-2}b\cdot dW - \frac{1}{2}\int |\sigma^{-1}b|^2 dt)$; it is the Itô–Stratonovich correction incurred when the stochastic integral in that exponent is rewritten as a pathwise (Stratonovich) functional evaluable on the smooth reference paths that the tube-limit computation uses, $\int b\cdot dW = \int b \circ dW - \frac{1}{2}\int \nabla\cdot b\, dt$ for $\sigma = I$; for gradient drift $b = -\nabla V$ it is $-\frac{1}{2}\int \Delta V\, dt$, the term produced by Itô's formula applied to V.

Freidlin–Wentzell rate functional Replacing $\sigma \to \varepsilon\sigma$ produces the small-noise family X^ε. Freidlin–Wentzell [29] prove the large deviation principle on $C([0,T];\mathbb{R}^d)$ with rate ε^2 and good rate functional $I[\xi] = \frac{1}{2}\int_0^T |\sigma^{-1}(\dot\xi - b(\xi))|^2\, dt$, i.e. the kinetic part of S^{OM} [67]. The minimum-action method [26] and the geometric minimum-action method [38] use this functional as a numerical tool; both start from (b,σ).

2.1 Indivisible targets and the no-input proposition

Following [70], an *indivisible target* is a two-time family $\{\Gamma(t \leftarrow s)\}_{0\le s\le t\le T}$ of column-stochastic kernels on Ω with $\Gamma(s \leftarrow s) = \mathrm{id}$ that satisfies (3). The *deficit operator* at a triple $0 \le s < r < t \le T$ is

$$\Delta(t, s, r) \;=\; \Gamma(t \leftarrow s) \;-\; \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s), \tag{5}$$

with operator norm $D(t, s, r) := \|\Delta(t, s, r)\|_{\mathrm{op}}$ on $L^2(\Omega, \mu)$, under the boundedness conventions of [70, §3]. Indivisibility is $D(t, s, r) > 0$ at some triple. By [70, Proposition 3.6], an indivisible family admits no strongly measurable, Bochner-integrable family of bounded generators reproducing it through a time-ordered exponential.

Proposition 2.1 (No input for the classical roles)**.** *Let* $\{\Gamma(t \leftarrow s)\}$ *be an indivisible target on* $\Omega \subseteq \mathbb{R}^d$*. Then there exists no drift–diffusion pair* (b, σ) *as in* (1) *whose diffusion has two-time transition kernels equal to* $\Gamma(t \leftarrow s)$*. So neither the tube-density role* (4) *nor the Freidlin–Wentzell rate-functional role of the OM action* in its drift–diffusion form (2) *is available for the target.*

Proof. An Itô diffusion is a Markov process, and the two-time transition kernels of a Markov process satisfy the Chapman–Kolmogorov factorization [70, Proposition 2.6]. A pair (b, σ) reproducing the target's kernels would therefore force $\Delta \equiv 0$, contradicting indivisibility. □

The conclusion is limited to what the proof uses. A non-Markovian target with a well-defined path law may possess its own small-tube asymptotics or its own large-deviation rate functional; Proposition 2.1 removes only the classical form derived from a pair (b, σ) reproducing the target's kernels.

Remark 2.2 (Scope of the exclusion). Proposition 2.1 excludes what its proof uses: a *Markov diffusion on the observed state space* Ω with the given two-time kernels. Three successively stronger exclusions must be kept distinct:

(E1) no generator on the observed state space whose evolution reproduces the given two-time family;

(E2) no finite hidden-Markov enlargement whose compression reproduces the family;

(E3) no Markovian enlargement of any kind.

Only (E1) follows from (3), and it is all this paper claims. (E2) is *false* for every process-induced family on a finite grid over a finite state space: the chain whose state is the path up to the present time is Markov on a finite state space and compresses to the given family [70, Remark 2.8]. (E3) does not follow either. Likewise, CK failure does not exclude a path-space law: a process-induced indivisible family has a well-defined law on path space, and CK failure removes only the generator-based derivation of a Girsanov-type path density from the observed two-time data. The converse direction is also delicate—two-time CK does not imply Markovianity (Feller's construction [27]; [70, Proposition 2.6(b) and Corollary 2.7])—so the two-time kernels neither pin down the path law nor certify a generator beyond (E1).

2.2 Degeneracy of the pointwise continuous-path weight

The natural continuous-path definition of a selection weight sets, for a surrogate pair (b_ν, σ_ν),

$$w_\nu(\xi) \;=\; \exp\!\big(-\lambda\, S_\nu^{\mathrm{OM}}[\xi]\big) \;\text{ for absolutely continuous } \xi, \qquad w_\nu(\xi) \;=\; 0 \;\text{ otherwise,} \tag{6}$$

with S_ν^{OM} the functional (2) evaluated at (b_ν, σ_ν). This pointwise weight is degenerate for exactly the proposals the framework needs.

Proposition 2.3 (Pointwise continuous-path weights vanish almost surely)**.** *Let* Π *be any probability measure on* $C([0,T];\mathbb{R}^d)$ *that assigns probability zero to the absolutely continuous paths. Then the weight* (6) *satisfies* $w_\nu = 0$ Π*-almost surely, the normalizing constant* $Z_\nu = \mathbb{E}_\Pi[w_\nu]$ *vanishes, and the self-normalized estimator is undefined* $(0/0)$ *for every sample size. The hypothesis holds for every proposal in the intended class: (i) the law of* σW *for a Brownian motion* W*; (ii) the Brownian bridge; (iii) the Ornstein–Uhlenbeck process from any initial distribution; and (iv) any Itô diffusion* (1) *with non-degenerate constant* σ *whose drift satisfies a Girsanov (Novikov-type) condition—in particular, any bounded drift—so that, conditionally on the initial value* $X_0 = x$*, its law on* $C([0,T];\mathbb{R}^d)$ *is equivalent to that of* $x + \sigma W$*.*

Proof. The conclusions are immediate from (6) once Π assigns probability zero to the absolutely continuous paths, so it suffices to verify the hypothesis in the four cases. (i) Brownian sample paths are almost surely nowhere differentiable, hence not absolutely continuous on any subinterval [57, Theorem 1.30]. (ii) The bridge $B_t = W_t - (t/T)\,W_T$ differs from a Brownian path by the random linear function $t \mapsto (t/T)\,W_T$; adding or subtracting a smooth path preserves absolute continuity in both directions, so the bridge's absolutely continuous paths form a null set by (i). (iii) By the Doob representation, the OU process $dX_t = -k(X_t - \bar{x})\,dt + \sigma\,dW_t$ with $k > 0$ satisfies $X_t - \bar{x} = e^{-kt}\big(X_0 - \bar{x}\big) + \sigma\,e^{-kt}\,\widetilde{W}\big(\tau(t)\big)$ with $\tau(t) = (e^{2kt} - 1)/(2k)$ and $\widetilde{W}$ a Brownian motion. Since $t \mapsto e^{-kt}$ is smooth and bounded away from zero on $[0,T]$ and τ is a smooth strictly increasing time change, X is absolutely continuous exactly when $\widetilde{W} \circ \tau$ is, hence exactly when $\widetilde{W}$ is absolutely continuous on $[0,\tau(T)]$—a null event by (i). (iv) The absolutely continuous paths form a Borel set that is invariant under constant shifts and null for the law of σW by (i), hence null for the law of $x + \sigma W$ for every x; conditionally on $X_0 = x$, Girsanov's theorem [44, Chapter IV] makes the diffusion's law equivalent to that of $x + \sigma W$ under the stated drift condition, so the conditional laws—and, integrating over the initial law, the unconditional law—assign the absolutely continuous paths probability zero. □

Proposition 2.3 is a statement about the pointwise weight (6), not about continuous-path inference. The classical OM functional is evaluated on regular reference paths through tube probabilities, as in (4), and a Girsanov change of measure is defined on path space without assigning an action to any individual sampled path; neither object is obstructed. The proposition invalidates the use of the classical functional as an ordinary pointwise importance weight on diffusion samples, which is the natural continuous-path version of the present construction; that version does not exist. In the Gaussian example of §9, the Girsanov ratio p_ν/p_Π of the surrogate's and the proposal's path densities on the same grids has chi-square divergence 0.1645, 0.1601, 0.1580, 0.1569, 0.1564, 0.1562 at $M = 25, 50, 100, 200, 400, 800$—bounded, and converging to a finite continuum value—while the selection weight's divergence grows to 9.8×10^{49} over the same range (§9.4). The repair is therefore structural: the object on which the estimator operates is a finite time grid, where the discrete action is a finite sum and the weights are strictly positive. The remainder of the paper is formulated on the grid; several grids appear only in the comparative refinement studies (§7.5, §9.4, §11) and in Proposition 7.5, each of which evaluates the same fixed-grid closed forms along a sequence of grids, and which together show why the importance-sampling representation of the selected measure has no continuum limit.

3 Finite-grid selection functionals and the estimator

3.1 Grid path space and observables

Fix a time grid $0 = t_0 < t_1 < \cdots < t_M = T$; for notational simplicity we take it uniform with step $\delta t = T/M$, all statements holding verbatim for non-uniform grids. The *grid path space* is

$$\mathcal{X}_M \;:=\; \Omega^{M+1} \;\subseteq\; \mathbb{R}^{d(M+1)}, \qquad \xi = (\xi_0, \xi_1, \ldots, \xi_M),$$

with its Borel σ-algebra. A *path observable* is a measurable function $F : \mathcal{X}_M \to \mathbb{R}$; the *terminal observables* are those of the form $F_\phi(\xi) = \phi(\xi_M)$ for measurable $\phi : \Omega \to \mathbb{R}$. We keep the typing strict throughout: paths are points of $\mathcal{X}_M$; observables are functions on $\mathcal{X}_M$; densities on Ω, where they appear (the initial density ρ_0 below, and the operator statements of §7), are elements of $L^1(\Omega, \mu)$ or $L^2(\Omega, \mu)$ and are never identified with paths or path functionals.

3.2 The proposal mechanism

Definition 3.1 (Proposal mechanism)**.** A *proposal mechanism* is a probability measure Π_M on $\mathcal{X}_M$ together with an i.i.d. sampling rule producing $\xi^{(1)}, \ldots, \xi^{(N)} \sim \Pi_M$.

Assumption 3.2 (Proposal conditions)**.** We impose the following on Π_M, flagging where each is used.

(P1) *Initial condition.* The law of ξ_0 under Π_M is a prescribed initial distribution on Ω: either a density ρ_0 with respect to μ, or a Dirac mass at a point $x_0 \in \Omega$ (as in the worked examples). Used for bookkeeping and for Theorem 7.1, where the target law is conditioned on the same initial distribution.

(P2) *Finite chi-square.* For every ν in a fixed compact $\Theta_0 \subset \Theta$, $\chi^2(Q_{\nu,M} \,\|\, \Pi_M) < \infty$, where $Q_{\nu,M}$ is the selected measure of Definition 3.8. Used for the central limit theorem, the variance theory, and Tier I of Theorem B; strong consistency needs less (Theorem 4.1(i)). By Lemma 3.9, (P2) holds automatically for surrogates whose drift divergence is bounded below, and Lemma 8.1 gives a checkable sufficient condition for polynomial drifts under Gaussian proposals.

Condition (P2) is the requirement under which the SNIS estimator has finite asymptotic variance uniformly over bounded observables [61, 2, 66]; it is strictly stronger than finiteness of the Kullback–Leibler divergence (Theorem 6.1(iii)) and strictly weaker than uniform boundedness of the weight ratio. The estimator needs only samples from Π_M; the closed-form and quadrature benchmarks of §9 additionally use the proposal's transition density, and that requirement is stated there, locally. Sample paths from standard candidate-generation procedures (forward-model simulators, neural-SDE generators, or bridge constructions), read off on the grid, yield proposal mechanisms satisfying Definition 3.1, provided each path is produced by an independent run; within-run particle ensembles from sequential Monte Carlo are dependent through resampling and belong to the adaptive extension discussed in §11.

3.3 The surrogate and the discrete selection action

Definition 3.3 (Surrogate)**.** A *surrogate* is a parameterized family $\{(b_\nu, \sigma_\nu)\}_{\nu \in \Theta}$, $\Theta \subseteq \mathbb{R}^p$ open, with $b_\nu \in C^1(\mathbb{R}^d; \mathbb{R}^d)$ and constant non-degenerate $\sigma_\nu \in \mathbb{R}^{d \times d}$, $\sigma_\nu \sigma_\nu^\top \succeq c\, I$ with $c > 0$ locally

uniformly in ν, and $\nu \mapsto (b_\nu, \sigma_\nu)$ continuous in C^1 on compact sets and in matrix norm. The *discrete selection action* of a grid path $\xi \in \mathcal{X}_M$ at parameter ν is

$$S_{\nu,M}[\xi] \;=\; \underbrace{\frac{1}{2}\sum_{m=0}^{M-1} \frac{\left|\sigma_\nu^{-1}\big(\xi_{m+1} - \xi_m - b_\nu(\xi_m)\,\delta t\big)\right|^2}{\delta t}}_{=:S^{\mathrm{kin}}_{\nu,M}[\xi]} \;+\; \underbrace{\frac{1}{2}\sum_{m=0}^{M-1} (\nabla\cdot b_\nu)(\xi_m)\,\delta t}_{=:S^{\mathrm{div}}_{\nu,M}[\xi]}. \tag{7}$$

Formula (7) is the left-point (Euler) discretization of the classical functional (2): the kinetic sum $S^{\mathrm{kin}}_{\nu,M}$ discretizes the kinetic term of (2) and the divergence sum $S^{\mathrm{div}}_{\nu,M}$ discretizes the Itô–Stratonovich correction. It is *not* the negative logarithm of the surrogate's Euler–Maruyama transition density: that density has $S^{\mathrm{kin}}_{\nu,M}$ as its exponent but carries a Gaussian normalization and no divergence term, and Proposition 3.5 makes the relation exact. On the grid (7) is a finite sum of finite terms, defined for *every* $\xi \in \mathcal{X}_M$ with no absolute-continuity proviso; the grid formulation is adopted for this reason, given Proposition 2.3. No continuum limit of (7) is invoked in this paper. The other standard convention—evaluating the drift at the midpoint of each increment—defines a different functional and, through Definition 3.8, a different estimand; the convention is part of the model (§9.6).

Remark 3.4 (The surrogate is not the truth). The surrogate is *not* interpreted as an approximation to the target. By Proposition 2.1, no member of any surrogate family is the true dynamics on the observed state space, and Proposition 7.2 bounds every divisible surrogate family away from an indivisible target's kernels by a deficit-determined threshold. The surrogate is a tractable computational object whose only role is to define the functional (7) on candidate paths.

3.4 The product identity

The following identity is elementary, and everything distinctive in the paper's behavior follows from it.

Proposition 3.5 (Product identity)**.** *Let $p_\nu(\xi)$ denote the surrogate's Euler transition density along the grid path ξ, conditionally on ξ_0, with respect to Lebesgue measure on Ω^M:*

$$p_\nu(\xi) \;=\; \prod_{m=0}^{M-1} (2\pi\,\delta t)^{-d/2}\,|\det\sigma_\nu|^{-1} \exp\Big(-\frac{|\sigma_\nu^{-1}(\xi_{m+1} - \xi_m - b_\nu(\xi_m)\delta t)|^2}{2\,\delta t}\Big).$$

Then for every $\xi \in \mathcal{X}_M$ and every $\lambda > 0$,

$$e^{-\lambda S_{\nu,M}[\xi]} \;=\; (2\pi\,\delta t)^{\lambda dM/2}\,|\det\sigma_\nu|^{\lambda M}\; p_\nu(\xi)^\lambda\; e^{-\lambda S^{\mathrm{div}}_{\nu,M}[\xi]}. \tag{8}$$

In particular, if the drift divergence is constant, $\nabla\cdot b_\nu \equiv -c_\nu$, then $e^{-\lambda S_{\nu,M}} = C_{\nu,M}\,p_\nu^\lambda$ with the constant $C_{\nu,M} = (2\pi\delta t)^{\lambda dM/2}|\det\sigma_\nu|^{\lambda M} e^{\lambda c_\nu T/2}$, and the selected measure of Definition 3.8 is

$$dQ_{\nu,M} \;\propto\; p_\nu^\lambda\, d\Pi_M. \tag{9}$$

Proof. Taking logarithms of the product defining p_ν gives $-\log p_\nu = S^{\mathrm{kin}}_{\nu,M} + \frac{dM}{2}\log(2\pi\delta t) + M\log|\det\sigma_\nu|$, so $e^{-\lambda S^{\mathrm{kin}}_{\nu,M}} = (2\pi\delta t)^{\lambda dM/2}|\det\sigma_\nu|^{\lambda M} p_\nu^\lambda$; multiply by $e^{-\lambda S^{\mathrm{div}}_{\nu,M}}$. For constant divergence, $S^{\mathrm{div}}_{\nu,M} \equiv -c_\nu M\delta t/2 = -c_\nu T/2$. □

For constant-divergence surrogates the selection weight is therefore the surrogate's own path density, tempered by λ, and the selected measure is the product of the proposal law and that tempered density, a product of experts on path space [39]. Every path-reweighting method we are aware of (§10.1) forms instead a *ratio* $p_{\text{target}}/p_{\text{reference}}$ of two path densities with a common diffusion coefficient, equivalently a difference of two actions; its weights are likelihood ratios. The present weight is a single density, and the difference in behavior follows from that. The OM divergence term $S^{\text{div}}_{\nu,M}$ distinguishes the selection action from the Euler log-density; for constant divergence it is a constant and cancels in the self-normalized ratio (13), so in that case the construction is exactly (9). It is exercised only by surrogates with non-constant divergence, and the second worked example (§9.5) shows that it then changes the estimand.

Corollary 3.6 (Gaussian pairs: precisions add)**.** *Let $d = 1$, let Π_M be the Euler chain $\xi_{m+1} = (1-k_\Pi\delta t)\xi_m + \sigma_\Pi\sqrt{\delta t}\,\eta_m$ started at $\xi_0 = 0$, let the surrogate be Ornstein–Uhlenbeck with (k_ν, σ_ν) and $\bar{x} = 0$, and assume $k_\Pi\delta t < 1$ and $k_\nu\delta t < 1$. Write K_Π and K_ν for the tridiagonal precision matrices on $(\xi_1, \dots, \xi_M)$ of the quadratic forms $\sum_m(\xi_{m+1} - a_\Pi\xi_m)^2/(\sigma_\Pi^2\delta t)$ and $\sum_m(\xi_{m+1} - a_\nu\xi_m)^2/(\sigma_\nu^2\delta t)$, $a_\bullet = 1 - k_\bullet\delta t$. Then:*

(i) $Q_{\nu,M} = \mathcal{N}\big(0, (K_\Pi + \lambda K_\nu)^{-1}\big)$; its precision is tridiagonal, so $Q_{\nu,M}$ is the law of a Gaussian Markov chain on the grid. (For an arbitrary centered Gaussian proposal $\mathcal{N}(0, K_\Pi^{-1})$ the first statement holds verbatim; the Markov property is inherited from the proposal's, not created by the tilt.)

(ii) The interior rows of $K_\Pi + \lambda K_\nu$ coincide exactly with those of the precision of the Euler chain of an Ornstein–Uhlenbeck process with parameters $(k_{\text{eff}}, \sigma_{\text{eff}})$ determined by

$$\frac{k_{\text{eff}}^2}{\sigma_{\text{eff}}^2} \;=\; \frac{k_\Pi^2}{\sigma_\Pi^2} + \lambda\,\frac{k_\nu^2}{\sigma_\nu^2}, \qquad \frac{1}{\sigma_{\text{eff}}^2} \;=\; \frac{1}{\sigma_\Pi^2} + \frac{\lambda}{\sigma_\nu^2} \;+\; O(\delta t); \tag{10}$$

only the last row differs, by an endpoint term (the first row coincides with the interior rows because $\xi_0 = 0$ is fixed). In particular the selected measure and the proposal have different quadratic variation for every $\lambda > 0$: $\sigma_{\text{eff}} < \sigma_\Pi$.

Proof. (i) Conditionally on $\xi_0 = 0$, Π_M has density $\propto e^{-\frac{1}{2}\xi^\top K_\Pi \xi}$ and, by Proposition 3.5, $w_\nu \propto e^{-\frac{\lambda}{2}\xi^\top K_\nu \xi}$; the product is Gaussian with precision $K_\Pi + \lambda K_\nu$. A Gaussian vector with tridiagonal precision has the property that each coordinate is conditionally independent of the rest given its two neighbors, which is the Markov property of the sequence. (ii) An interior row of the precision of an Euler chain with regression coefficient a and innovation variance $s^2\delta t$ has diagonal entry $(1 + a^2)/(s^2\delta t)$ and off-diagonal entries $-a/(s^2\delta t)$. For $K_\Pi + \lambda K_\nu$ the corresponding entries are $v/\delta t$ and $-u/\delta t$ with $u = a_\Pi/\sigma_\Pi^2 + \lambda a_\nu/\sigma_\nu^2 > 0$ and $v = (1 + a_\Pi^2)/\sigma_\Pi^2 + \lambda(1 + a_\nu^2)/\sigma_\nu^2$. Matching coefficients, $a/s^2 = u$ and $(1+a^2)/s^2 = v$; since $1+a_\bullet^2 \geq 2a_\bullet$ one has $v \geq 2u$, so $a^2 - (v/u)a + 1 = 0$ has a root $a_{\text{eff}} \in (0, 1]$, and $\sigma_{\text{eff}}^2 = a_{\text{eff}}/u$. Writing $a_{\text{eff}} = 1 - k_{\text{eff}}\delta t$ and substituting $a_\bullet = 1 - k_\bullet\delta t$, the identity $v - 2u = k_{\text{eff}}^2\delta t^2/\sigma_{\text{eff}}^2$ evaluates exactly to $\delta t^2(k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2)$, which is the first relation in (10); and $1/\sigma_{\text{eff}}^2 = u/(1 - k_{\text{eff}}\delta t) = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$, which is the second. That $\sigma_{\text{eff}} < \sigma_\Pi$ holds for every $\lambda > 0$, and not only when $\lambda \gg \delta t$, follows from the Toeplitz structure: the interior rows of K_Π, K_ν and $K_\Pi + \lambda K_\nu$ are those of stationary autoregressions with symbols $\hat{K}_\Pi(\omega) = (1 + a_\Pi^2 - 2a_\Pi\cos\omega)/(\sigma_\Pi^2\delta t)$, $\hat{K}_\nu(\omega)$ likewise, and $\hat{K}_\Pi + \lambda\hat{K}_\nu$; the innovation variance of a stationary Gaussian autoregression is $\exp\big(-\frac{1}{2\pi}\int_{-\pi}^{\pi}\log\hat{K}(\omega)\,d\omega\big)$ (the Kolmogorov–Szegő formula [10, §5.8]), which is strictly decreasing in the symbol, and $\hat{K}_\nu > 0$ pointwise. □

Remark 3.7 (Consequences of the identity). Three things follow and are used throughout. (a) *The temperature is a diffusion-coefficient control.* By (10), changing λ changes the quadratic variation of the selected measure, not only the strength of a reweighting; different temperatures define different selected measures, and λ is a modeling parameter with no selection principle offered here beyond the diagnostics of Theorem C. (b) *The selected measure and the proposal are mutually singular in the continuum.* Diffusion laws with different diffusion coefficients are mutually singular; on the grid this appears as chi-square divergence growing exponentially in M at fixed λ, reported in §9.4; the refinement study of the July version[1] attributed it instead to Proposition 2.3. Since $\sigma_{\text{eff}} < \sigma_\Pi$ for every $\lambda > 0$, the degeneracy is not removable by any choice of temperature. (c) *The selected measure may be Markov.* For a Gaussian Markov proposal it is (Corollary 3.6(i)); for a Gaussian proposal with longer-range dependence it inherits that dependence, as in Example 7.4. Indivisibility of the target excludes only that $Q_{\nu,M}$ be the law of a generator-based model *reproducing the target family*, not that it be Markov.

3.5 Selection weight, selected measure, and estimator

Definition 3.8 (Selection weight and selected measure)**.** Fix a temperature $\lambda > 0$. The *selection weight* of $\xi \in \mathcal{X}_M$ at parameter ν is

$$w_\nu(\xi) \;=\; \exp\bigl(-\lambda\, S_{\nu,M}[\xi]\bigr) \;\in\; (0,\infty), \tag{11}$$

and, whenever $Z_\nu := \mathbb{E}_{\Pi_M}[w_\nu] \in (0,\infty)$, the *selected measure* is the Gibbs reweighting

$$dQ_{\nu,M} \;=\; \frac{w_\nu}{Z_\nu}\, d\Pi_M, \qquad \rho_\nu := \frac{dQ_{\nu,M}}{d\Pi_M} = \frac{w_\nu}{Z_\nu}. \tag{12}$$

Lemma 3.9 (Well-posedness on the grid)**.** *For every $\xi \in \mathcal{X}_M$ and $\nu \in \Theta$, $0 < w_\nu(\xi) < \infty$. If moreover the drift divergence is bounded below, $(\nabla\cdot b_\nu)(x) \geq -c_\nabla(\nu)$ for all $x \in \Omega$, then*

$$0 \;<\; Z_\nu \;\leq\; W_{\max}(\nu) := \exp\bigl(\lambda\, c_\nabla(\nu)\, T/2\bigr) \;<\; \infty,$$

$$\chi^2(Q_{\nu,M} \,\|\, \Pi_M) \;=\; \frac{\mathbb{E}_{\Pi_M}[w_\nu^2]}{Z_\nu^2} - 1 \;\leq\; \frac{W_{\max}(\nu)}{Z_\nu} - 1 \;<\; \infty,$$

so the selected measure exists and Assumption 3.2(P2) holds automatically. Without the divergence bound, $Z_\nu > 0$ still holds whenever $Z_\nu < \infty$, and (P2) is a genuine assumption.

Proof. $S_{\nu,M}[\xi]$ is a finite sum of finite terms, so $w_\nu(\xi) \in (0,\infty)$ pointwise; positivity of Z_ν follows from $w_\nu > 0$ and Π_M being a probability measure. Under the divergence bound, the kinetic sum in (7) is non-negative and the divergence sum is at least $-c_\nabla T/2$, so $S_{\nu,M} \geq -c_\nabla T/2$ and $0 < w_\nu \leq W_{\max}$; then $w_\nu^2 \leq W_{\max} w_\nu$ pointwise, so $\mathbb{E}[w_\nu^2] \leq W_{\max} Z_\nu$, which is the displayed chi-square bound. □

For the Ornstein–Uhlenbeck surrogate in dimension d (§8.1), $\nabla \cdot b_\nu = -kd$ is constant, so $W_{\max} = e^{\lambda k T d/2}$ exactly; for polynomial drifts of degree ≥ 2 the divergence is in general unbounded below and (P2) must be certified for the proposal at hand, which Lemma 8.1 does for Gaussian proposals. Boundedness of the weights does *not* make the weight distribution benign, and the chi-square bound of the lemma certifies finiteness and nothing more: the Gaussian example has $W_{\max} = e$ and $W_{\max}/Z_\nu \approx 3.5 \times 10^7$, so the lemma's bound reads $\chi^2 \leq 3.5 \times 10^7$ against the exact value 1372 (§9.2); this is exactly the heavy-tailed-ratio regime the diagnostics of Theorem C are for.

[1] The July 2026 deposit, SSRN DOI 10.2139/ssrn.6922758.

Definition 3.10 (Selection-functional estimator)**.** For a path observable $F : \mathcal{X}_M \to \mathbb{R}$ and an i.i.d. sample $\xi^{(1)}, \dots, \xi^{(N)} \sim \Pi_M$, the *selection-functional estimator* of F at parameter ν is

$$\hat{F}_N(\nu) \;=\; \frac{\sum_{i=1}^N F(\xi^{(i)})\, w_\nu(\xi^{(i)})}{\sum_{i=1}^N w_\nu(\xi^{(i)})}. \tag{13}$$

The ratio (13) is a self-normalized importance-sampling estimator with proposal Π_M and importance-sampling target $Q_{\nu,M}$, and Theorem A identifies its limit as $F^*(\nu) = \mathbb{E}_{Q_{\nu,M}}[F]$: the estimator consistently evaluates observables under the path measure selected jointly by the proposal and the surrogate action. Under (3) the unnormalized measure $w_\nu\, d\Pi_M$ is not established as the path-space law of any generator-based model *reproducing the target family*, and Z_ν has no interpretation as a marginal likelihood of the target; the selected measure is nonetheless a well-defined product measure, and may itself be Markov (Corollary 3.6). The estimand $F^*(\nu)$ depends on the proposal, the surrogate, the temperature, the grid and the discretization convention (§9.6); its relation to the target process is the content of §7.

Remark 3.11 (Generalized-Bayes form)*.* Treating $S_{\nu,M}$ as a loss and λ as a learning rate, the unnormalized measure $w_\nu\, d\Pi_M$ has the algebraic *form* of a Bissiri–Holmes–Walker generalized-Bayes update [7, 40, 48]. The analogy is formal only. In a generalized-Bayes update the prior is over a parameter and the loss is evaluated at observed data that condition it; here the candidate path is itself the variable distributed under $Q_{\nu,M}$, and no data condition it—observed data enter only through the calibration of ν (§8.4), a separate operation. $Q_{\nu,M}$ is a Gibbs measure on candidate paths, a product of experts by Proposition 3.5, and not a posterior in the inferential sense.

Remark 3.12 (Causal factorization)*.* The action (7) is adapted: its restriction to the first m increments depends only on $(\xi_0, \dots, \xi_m)$. Then the weight factorizes causally, $w_\nu(\xi) = \prod_{m=0}^{M-1} w_\nu^{(m)}(\xi_m, \xi_{m+1})$ with per-increment factors $w_\nu^{(m)} = \exp(-\lambda[\,|\sigma_\nu^{-1}(\xi_{m+1} - \xi_m - b_\nu(\xi_m)\delta t)|^2/(2\delta t) + (\nabla\cdot b_\nu)(\xi_m)\delta t/2\,])$. Sequential (particle-filter) implementations can exploit this factorization when the proposal itself admits compatible sequential sampling—a Markov Π_M, or one that factorizes causally in the same way; the factorization of the weight alone does not license a particle filter for an arbitrary proposal, which Definition 3.1 only requires to be samplable as whole paths.

4 Theorem A: consistency and asymptotic normality

Throughout this section and the next two, Π_M, w_ν, Z_ν, $Q_{\nu,M}$, and $\rho_\nu = w_\nu/Z_\nu$ are as in §3, the parameter $\nu \in \Theta_0$ is fixed, and

$$A_N \;=\; \frac{1}{N}\sum_{i=1}^N F(\xi^{(i)})\, w_\nu(\xi^{(i)}), \qquad B_N \;=\; \frac{1}{N}\sum_{i=1}^N w_\nu(\xi^{(i)}), \qquad \hat{F}_N(\nu) \;=\; \frac{A_N}{B_N}.$$

The three results of this section are the standard self-normalized importance-sampling limit theorems [30, 33, 61] specialized to the selection weight; they are stated in full because the later sections use their hypotheses.

Theorem 4.1 (A: strong consistency, central limit theorem, asymptotic mean-squared error)**.** *Let $F : \mathcal{X}_M \to \mathbb{R}$ be measurable and suppose $Z_\nu \in (0, \infty)$.*

*(i) (*Strong consistency*) If $\mathbb{E}_{\Pi_M}|F\, w_\nu| < \infty$, then*

$$\hat{F}_N(\nu) \;\xrightarrow{a.s.}\; F^*(\nu) \;:=\; \frac{\mathbb{E}_{\Pi_M}[F\, w_\nu]}{Z_\nu} \;=\; \mathbb{E}_{Q_{\nu,M}}[F] \qquad \text{as } N \to \infty. \tag{14}$$

(ii) (Central limit theorem, CLT) *If moreover* $\mathbb{E}_{\Pi_M}[w_\nu^2] < \infty$ *and* $\mathbb{E}_{\Pi_M}[F^2 w_\nu^2] < \infty$, *then*

$$\sqrt{N}\big(\hat{F}_N(\nu) - F^*(\nu)\big) \xrightarrow{d} \mathcal{N}\big(0, \sigma^2(\nu)\big), \qquad \sigma^2(\nu) = \frac{\mathbb{E}_{\Pi_M}\big[(F - F^*(\nu))^2\, w_\nu^2\big]}{Z_\nu^2}. \tag{15}$$

(iii) (Asymptotic mean-squared error) *If, in addition to the hypotheses of (ii), the family* $\{N(\hat{F}_N(\nu) - F^*(\nu))^2\}_{N \geq 1}$ *is uniformly integrable, then*

$$N\,\mathbb{E}\big[(\hat{F}_N(\nu) - F^*(\nu))^2\big] \longrightarrow \sigma^2(\nu) \qquad \text{as } N \to \infty.$$

Proof. (i) The strong law of large numbers [30] gives $A_N \to \mathbb{E}_{\Pi_M}[F w_\nu]$ and $B_N \to Z_\nu > 0$ almost surely; the continuous mapping theorem applied to the ratio gives (14). (ii) Set $X_i = (F(\xi^{(i)}) - F^*)\, w_\nu(\xi^{(i)})$, which are i.i.d. with $\mathbb{E}X_i = \mathbb{E}_{\Pi_M}[F w_\nu] - F^* Z_\nu = 0$ and $\mathrm{Var}(X_i) = \tilde{\sigma}^2 := \mathbb{E}_{\Pi_M}[(F - F^*)^2 w_\nu^2] < \infty$. Then $\hat{F}_N - F^* = (N^{-1}\sum_i X_i)/B_N$; the classical CLT gives $N^{-1/2}\sum_i X_i \Rightarrow \mathcal{N}(0, \tilde{\sigma}^2)$, $B_N \to Z_\nu$ a.s., and Slutsky's theorem yields (15) with $\sigma^2 = \tilde{\sigma}^2 / Z_\nu^2$; this is the standard SNIS asymptotic variance [30, 33, 61], equivalently the delta method applied to $(A_N, B_N) \mapsto A_N / B_N$. (iii) By (ii) and the continuous mapping theorem, $N(\hat{F}_N - F^*)^2 \Rightarrow \sigma^2(\nu)\,\zeta^2$ with $\zeta \sim \mathcal{N}(0,1)$; uniform integrability upgrades convergence in distribution to convergence of expectations. □

The moment hypotheses in (ii) are standard sufficient conditions, not the weakest possible: integrability of the denominator together with finite variance of $(F - F^*)w_\nu$ suffices for the argument as written.

Remark 4.2 (Part (iii) is conditional). No primitive sufficient condition for the uniform integrability of $\{N(\hat{F}_N - F^*)^2\}$ is supplied, and in the heavy-tailed-weight regime that the diagnostics of Theorem C describe it is not a formality: uniform integrability can fail while (ii) holds, in which case the central limit theorem describes the distributional limit but the scaled mean-squared error need not converge to $\sigma^2(\nu)$. The asymptotic mean-squared-error statement is therefore unavailable in practice unless such a condition is established separately for the pair at hand. Theorem B gives finite-N mean-squared-error statements.

Remark 4.3 (No finite-N mean-squared-error identity from second moments). Second moments alone do not yield an exact finite-N bound of the form $\mathbb{E}[(\hat{F}_N - F^*)^2] \leq V/N$: the denominator B_N is random, and on the event that it is small the ratio is uncontrolled by variances. The asymptotic variance also does not factor as $(1 + \chi^2)\,\mathrm{Var}_{Q_{\nu,M}}(F)$ in general: the correct identity is $\sigma^2(\nu) = \mathbb{E}_{Q_{\nu,M}}[(F - F^*)^2 \rho_\nu]$ (Theorem 6.1(i)), in which the squared deviation and the density ρ_ν are integrated *jointly*, and the product form holds exactly when $(F - F^*)^2$ and ρ_ν are uncorrelated under $Q_{\nu,M}$, which has no reason to hold. Genuine finite-N statements require either the polynomial bounds of Theorem 5.1 (finite chi-square) or the exponential bounds of Theorem 5.2 (bounded weights), both of which control the denominator event explicitly.

5 Theorem B: finite-sample concentration in two tiers

The two tiers are calibrated to the two weight regimes that occur (Lemma 3.9): finite chi-square with possibly unbounded weights (Tier I, polynomial concentration), and bounded weights (Tier II, exponential concentration). Finite variance does not imply exponential concentration—an importance weight may have a polynomial tail with finite variance—so the tiers are distinct. The inequalities used are classical [9]; the content of the section is the matching of tier to regime and the explicit control of the random denominator.

Theorem 5.1 (B–I: polynomial concentration under finite chi-square)**.** *Assume $Z_\nu \in (0,\infty)$, $\mathbb{E}_{\Pi_M}|F\,w_\nu| < \infty$ (so that $F^*(\nu)$ is defined), $\chi^2 := \chi^2(Q_{\nu,M}\|\Pi_M) < \infty$, and $\tilde\sigma^2 = \mathbb{E}_{\Pi_M}[(F - F^*)^2 w_\nu^2] < \infty$, and write $\sigma^2(\nu) = \tilde\sigma^2/Z_\nu^2$. Then for every $t > 0$ and every $N \geq 1$,*

$$\mathbb{P}\Big(\big|\hat F_N(\nu) - F^*(\nu)\big| > t\Big) \;\leq\; \frac{4\,\sigma^2(\nu)}{N\,t^2} \;+\; \frac{4\,\chi^2}{N}, \tag{16}$$

where the second term bounds the denominator event by the Cantelli (one-sided Chebyshev) inequality:

$$\mathbb{P}\Big(B_N < \tfrac{Z_\nu}{2}\Big) \;\leq\; \frac{\mathrm{Var}(w_\nu)/N}{\mathrm{Var}(w_\nu)/N + Z_\nu^2/4} \;\leq\; \frac{4\,\mathrm{Var}(w_\nu)}{N\,Z_\nu^2} \;=\; \frac{4\,\chi^2}{N}, \tag{17}$$

using $\mathrm{Var}(w_\nu) = Z_\nu^2\,\chi^2$. If moreover $|F| \leq F_{\max}$ Π_M-a.s., then the finite-N mean-squared-error bound

$$\mathbb{E}\big[(\hat F_N(\nu) - F^*(\nu))^2\big] \;\leq\; \frac{4\,\sigma^2(\nu)}{N} \;+\; \frac{16\,F_{\max}^2\,\chi^2}{N} \tag{18}$$

holds for every $N \geq 1$.

Proof. Let $X_i = (F(\xi^{(i)}) - F^*)w_\nu(\xi^{(i)})$ as in the proof of Theorem 4.1 and $S_N = \sum_{i=1}^N X_i$, so that on the event $\{B_N \geq Z_\nu/2\}$,

$$\big|\hat F_N - F^*\big| \;=\; \frac{|N^{-1}S_N|}{B_N} \;\leq\; \frac{2\,|N^{-1}S_N|}{Z_\nu}.$$

Chebyshev's inequality gives $\mathbb{P}(|N^{-1}S_N| > Z_\nu t/2) \leq 4\tilde\sigma^2/(NZ_\nu^2 t^2) = 4\sigma^2(\nu)/(Nt^2)$. For the denominator, B_N is a mean of i.i.d. variables with mean Z_ν and variance $\mathrm{Var}(w_\nu)/N$; Cantelli's inequality [9, Exercise 2.3] applied to the lower deviation $B_N - Z_\nu \leq -Z_\nu/2$ gives (17), and $\mathrm{Var}(w_\nu) = \mathbb{E}[w_\nu^2] - Z_\nu^2 = Z_\nu^2\chi^2$ exactly. The union of the two events gives (16). For (18): $\hat F_N$ is a convex combination of values of F, so $|\hat F_N| \leq F_{\max}$ and $|F^*| \leq F_{\max}$ almost surely; splitting the expectation on $\{B_N \geq Z_\nu/2\}$ and its complement,

$$\mathbb{E}\big[(\hat F_N - F^*)^2\big] \;\leq\; \frac{4}{Z_\nu^2}\,\mathbb{E}\big[(N^{-1}S_N)^2\big] \;+\; (2F_{\max})^2\,\mathbb{P}\Big(B_N < \tfrac{Z_\nu}{2}\Big) \;\leq\; \frac{4\tilde\sigma^2}{NZ_\nu^2} + \frac{16F_{\max}^2\chi^2}{N}. \qquad \square$$

The bounds (16) and (18) are of the correct order in N, but their constants are not sharp: when N is small relative to χ^2 the right-hand side of (16) exceeds one and that of (18) exceeds the trivial bound $4F_{\max}^2$, and they then carry no information. They are finite-N statements of asymptotic order, informative once $N \gg \chi^2$, and not uniformly informative guarantees.

Theorem 5.2 (B–II: exponential concentration under bounded weights)**.** *Assume, in addition to the hypotheses of Theorem 5.1, that $0 < w_\nu \leq W_{\max}$ and $|F| \leq F_{\max}$ Π_M-a.s. Then for every $t > 0$ and every $N \geq 1$,*

$$\mathbb{P}\Big(\big|\hat F_N(\nu) - F^*(\nu)\big| > t\Big) \;\leq\; 2\exp\!\left(-\frac{N\,t^2}{8\,\sigma^2(\nu) \;+\; \frac{8}{3}\,F_{\max}W_{\max}\,t/Z_\nu}\right) \;+\; \exp\!\left(-\frac{N\,Z_\nu^2}{2\,W_{\max}^2}\right), \tag{19}$$

the first term by Bernstein's inequality on the numerator sum and the second by Hoeffding's inequality on the denominator, with the $W_{\max}^2$ scaling dictated by the range of a single weight.

Proof. The summands $X_i = (F_i - F^*)w_i$ satisfy $|X_i| \leq 2F_{\max}W_{\max} =: c_b$ and $\mathrm{Var}(X_i) = \tilde\sigma^2$. Bernstein's inequality [9, Theorem 2.10] gives, for $u > 0$,

$$\mathbb{P}\big(|S_N| \geq Nu\big) \;\leq\; 2\exp\!\left(-\frac{Nu^2}{2\tilde\sigma^2 + \frac{2}{3}\,c_b\,u}\right).$$

Taking $u = Z_\nu t/2$ and simplifying with $\tilde{\sigma}^2 = Z_\nu^2 \sigma^2(\nu)$ yields the first term of (19) on the event $\{B_N \geq Z_\nu/2\}$. For the denominator, $w_\nu \in [0, W_{\max}]$ has range at most $W_{\max}$, so the one-sided Hoeffding inequality [9, Theorem 2.8] gives

$$\mathbb{P}\Big(B_N - Z_\nu \leq -\tfrac{Z_\nu}{2}\Big) \;\leq\; \exp\!\left(-\frac{2N(Z_\nu/2)^2}{W_{\max}^2}\right) \;=\; \exp\!\left(-\frac{N Z_\nu^2}{2W_{\max}^2}\right). \qquad \square$$

Corollary 5.3 (Hoeffding form)**.** *Under the hypotheses of Theorem 5.2, replacing Bernstein by Hoeffding on the numerator (summand range $4F_{\max}W_{\max}$),*

$$\mathbb{P}\Big(\big|\hat{F}_N(\nu) - F^*(\nu)\big| > t\Big) \;\leq\; 2\exp\!\left(-\frac{N\,t^2\,Z_\nu^2}{32\,F_{\max}^2\,W_{\max}^2}\right) \;+\; \exp\!\left(-\frac{N\,Z_\nu^2}{2\,W_{\max}^2}\right). \tag{20}$$

Remark 5.4 (Which tier binds). Tier II applies whenever the divergence bound of Lemma 3.9 holds, but its constants degrade with the ratio $W_{\max}/Z_\nu$: in the Gaussian example, $W_{\max} = e$ while $Z_\nu \approx 7.9 \times 10^{-8}$, so the denominator exponent in (19)–(20) is $\exp(-4.2 \times 10^{-16} N)$ and Tier II is vacuous below $N \sim 10^{15}$; the polynomial Tier I together with the chi-square diagnostics of Theorem C carries the operational content. The exponential tier is the least informative statement in the paper's own examples, because the product structure of Proposition 3.5 makes Z_ν the overlap of two path densities and hence exponentially small in M. Conversely, when the drift divergence is unbounded below, the automatic bound of Lemma 3.9 is unavailable; unless the proposal itself confines the paths, so that the Π_M-essential supremum of w_ν is finite, Tier I is the only finite-sample statement. Weights with sub-exponential (rather than bounded) tails admit Bernstein-type bounds with the corresponding variance and scale parameters [9, §2.8]; we do not state them separately. The general theory of self-normalized processes [63] provides exponential bounds for ratios of the form $S_N/\sqrt{\sum_i X_i^2}$ under its own hypotheses (symmetry or moment conditions on the self-normalized ratio); those hypotheses are not implied by finite chi-square, and we cite that theory for context rather than invoking it.

6 Theorem C: degeneracy diagnostics and sample-size scales

Theorem 6.1 (C: variance, effective sample size, and divergence ordering)**.** *Assume $Z_\nu \in (0, \infty)$ and $\chi^2 = \chi^2(Q_{\nu,M} \| \Pi_M) < \infty$, and let $\rho_\nu = w_\nu/Z_\nu$ be the selected-measure density* (12)*.*

*(i) (*Variance identity and bound*) The asymptotic variance of* (15) *satisfies*

$$\sigma^2(\nu) \;=\; \mathbb{E}_{Q_{\nu,M}}\big[(F - F^*(\nu))^2\,\rho_\nu\big], \qquad \mathbb{E}_{Q_{\nu,M}}[\rho_\nu] \;=\; 1 + \chi^2,$$

and hence $\sigma^2(\nu) \leq 4F_{\max}^2\,(1+\chi^2)$ whenever $|F| \leq F_{\max}$ Π_M-a.s.

*(ii) (*Effective sample size*) The plug-in effective sample size $\hat{N}_{\text{eff}} = \big(\sum_i w_\nu^{(i)}\big)^2 / \sum_i \big(w_\nu^{(i)}\big)^2$ satisfies*

$$\frac{\hat{N}_{\text{eff}}}{N} \;\xrightarrow{\;a.s.\;}\; \frac{1}{1+\chi^2} \qquad \text{as } N \to \infty.$$

*(iii) (*Divergence ordering*) Unconditionally,*

$$\exp\big(D_{\mathrm{KL}}(Q_{\nu,M} \,\|\, \Pi_M)\big) \;\leq\; 1 + \chi^2.$$

Proof. (i) $\sigma^2 = Z_\nu^{-2}\mathbb{E}_{\Pi_M}[(F-F^*)^2 w_\nu^2] = \mathbb{E}_{\Pi_M}[(F-F^*)^2\rho_\nu^2] = \mathbb{E}_{Q_{\nu,M}}[(F-F^*)^2\rho_\nu]$, and $\mathbb{E}_Q[\rho_\nu] = \mathbb{E}_{\Pi_M}[\rho_\nu^2] = \mathbb{E}[w^2]/Z^2 = 1+\chi^2$; the bound uses $|F-F^*| \le 2F_{\max}$. (ii) By the strong law, $B_N \to Z_\nu$ and $N^{-1}\sum_i w_i^2 \to \mathbb{E}[w^2]$ a.s.; hence $\hat{N}_{\text{eff}}/N = B_N^2/(N^{-1}\sum_i w_i^2) \to Z_\nu^2/\mathbb{E}[w^2] = 1/(1+\chi^2)$. (iii) By Jensen's inequality for the concave logarithm, $D_{\text{KL}}(Q\|\Pi) = \mathbb{E}_Q[\log\rho_\nu] \le \log\mathbb{E}_Q[\rho_\nu] = \log(1+\chi^2)$; this is the monotonicity of Rényi divergences in the order, $D_1 \le D_2$ [74]. □

Part (i) makes precise in what sense $1+\chi^2$ is a variance-inflation factor of the selection step: it is an observable-independent upper-bound factor, attained in order of magnitude when the deviation $(F-F^*)^2$ correlates with the density ρ_ν, and not the actual inflation for a particular F; the popular reading $N_{\text{eff}} = N/(1+\chi^2)$ [53, 61] is the corresponding worst-case deflator, not an identity for a fixed observable (Remark 4.3). Part (ii) states the sense in which the plug-in diagnostic is consistent. Its finite-sample behavior differs: the plug-in $\hat{\chi}^2$ and $\hat{N}_{\text{eff}}$ are themselves dominated by the upper tail of w_ν and are severely negatively biased exactly in the heavy-tailed regime, a bias the worked examples measure against the exact values (§9.4).

Sample-size scales as diagnostics Two published sample-size results attach to the divergences in Theorem 6.1(iii), and we state them as *diagnostic scales*, with their hypotheses, rather than as universal necessary conditions for a fixed observable. Chatterjee–Diaconis [15] show that, for importance sampling of $\mathbb{E}_Q[F]$ with $\|F\|_{L^2(Q)} \le 1$, sample sizes $N \ge \exp(D_{\text{KL}} + t)$ suffice for accuracy with failure probability of order $e^{-t/4}$, while for $N \le \exp(D_{\text{KL}} - t)$ there *exist* bounded observables for which the estimator fails; the scale $\exp(D_{\text{KL}})$ is thus sufficient uniformly over an $L^2(Q)$ ball and necessary for the class, not for each fixed F. Sanz-Alonso [66] shows that guarantees on the relative mean error uniformly over test-function classes require N at the scale $1+\chi^2$—again a class statement, with the constants depending on the class and the accuracy. The two scales are ordered by Theorem 6.1(iii), $\exp(D_{\text{KL}}) \le 1+\chi^2$, and they respond differently to the accuracy level, so neither dominates operationally in all regimes: when the weight ratio is heavy-tailed—χ^2 large relative to $\exp(D_{\text{KL}})$, which is the regime in which a reweighting object is needed—the chi-square scale is the binding diagnostic, while $\exp(D_{\text{KL}})$ remains the available reference when only the Kullback–Leibler divergence is computable. Both worked examples sit in the chi-square-binding regime ($\exp(D_{\text{KL}}) \approx 129$ against $1+\chi^2 \approx 1373$ for the Gaussian pair; 27 against 163 for the double well; §9).

Remark 6.2 (Diagnostics in practice are heuristics). $\hat{N}_{\text{eff}}$ and the plug-in $\hat{\chi}^2$ are computable at no additional cost during the SNIS sweep. Two operational rules of thumb are in common use and we record them as such, not as consequences of Theorem 6.1 or of anything else in this paper: when $\hat{N}_{\text{eff}} \lesssim 10^2$ the estimator is treated as unreliable for all but the most stable observables, and when $\hat{N}_{\text{eff}}/N \lesssim 10^{-2}$ Pareto-smoothed importance sampling [75] is often used as a tail stabilizer. Neither threshold follows from a theorem; their validity depends on the observable, the weight-tail behavior, and the accuracy required. The smoothing method carries its own diagnostic—the Pareto shape estimate $\hat{k}$ of Vehtari and coauthors, which is the quantity a practitioner following the recommendation computes—and it should be reported alongside $\hat{N}_{\text{eff}}$; and modifying the weights changes the estimator, so the smoothed estimator is not the one to which Theorems A–C apply. The negative bias of the plug-in diagnostics in the heavy-tailed regime is not a heuristic (Theorem 6.1(ii) describes only their limit) and should be assumed rather than ruled out.

7 Theorem D: the truth link as a conditional transfer, and the obstruction

Everything so far concerns the selected measure $Q_{\nu,M}$. This section states exactly what can be said about the target process: a conditional transfer under an explicit measure-level approximation hypothesis (Theorem 7.1); an obstruction theorem at the level of kernels (Proposition 7.2); an example showing that the two are logically independent (Example 7.4); and the design rule that governs when the transfer hypothesis can hold at all (§7.5), which is the form the open problem must take.

7.1 Two settings, and the grid law

Two settings must be kept apart, and every statement about the target is typed by one of them. In the *kernel setting* (K), only the two-time family $\{\Gamma(t \leftarrow s)\}$ on Ω is available, restricted to the grid $\{t_0, \dots, t_M\}$. Two-time kernels do not determine a joint law on $\mathcal{X}_M$ (Remark 2.2), so in (K) general path observables under the target are not identified; the target-side objects are the kernels themselves and functionals of them, such as the deficit $D(t, s, r)$. In the *process setting* (L), the family is *process-induced*: there exists a stochastic process $(X_{t_m})_{m=0}^{M}$ on Ω whose conditional transition kernels on the grid are the $\Gamma(t_j \leftarrow t_i)$, and that process is specified. Then its law $P_{\mathrm{true},M}$ on $\mathcal{X}_M$, conditioned on the initial distribution of Assumption 3.2(P1), is a fixed object, $F_{\mathrm{true}} = \mathbb{E}_{P_{\mathrm{true},M}}[F]$ is defined, and path-space comparisons and divergences are meaningful. Theorems A–C belong to neither setting: they concern $Q_{\nu,M}$ alone. Theorem 7.1 lives in (L); Proposition 7.2 lives in (K); the comparison table of §10 records the setting of each row.

7.2 The conditional transfer

Theorem 7.1 (D: conditional transfer)**.** *In setting (L), let $F : \mathcal{X}_M \to \mathbb{R}$ satisfy $|F| \le F_{\max}$, and let ν be such that $Z_\nu \in (0, \infty)$. Assume the* measure-level approximation hypothesis

(D1) $d_{\mathrm{TV}}\big(Q_{\nu,M},\, P_{\mathrm{true},M}\big) \;\le\; \eta_{\nu,M}$ *for a tolerance $\eta_{\nu,M} \ge 0$ established or estimated by means external to this paper,*

where $d_{\mathrm{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$. Then

$$\big|F^*(\nu) - F_{\mathrm{true}}\big| \;\le\; 2\, F_{\max}\, \eta_{\nu,M}, \tag{21}$$

and, by Theorems A and B, $\hat{F}_N(\nu) \to F^*(\nu)$ almost surely with the stated finite-sample deviations, so that the total error decomposes as

$$\big|\hat{F}_N(\nu) - F_{\mathrm{true}}\big| \;\le\; \underbrace{\big|\hat{F}_N(\nu) - F^*(\nu)\big|}_{\text{estimator error: Theorems A–C}} \;+\; \underbrace{2\, F_{\max}\, \eta_{\nu,M}}_{\text{selection bias: hypothesis (D1)}}.$$

For terminal observables $F_\phi(\xi) = \phi(\xi_M)$ with $|\phi| \le F_{\max}$, (D1) may be weakened to a total-variation bound on the time-t_M marginals alone.

Proof. We have

$$|F^*(\nu) - F_{\mathrm{true}}| = \left|\int F\, d(Q_{\nu,M} - P_{\mathrm{true},M})\right| \le F_{\max}\, \|Q_{\nu,M} - P_{\mathrm{true},M}\|_{\mathrm{TV}} = 2 F_{\max}\, d_{\mathrm{TV}}(Q_{\nu,M}, P_{\mathrm{true},M}),$$

and for terminal observables the signed measure may be replaced by the difference of the terminal marginals. □

Theorem 7.1 transfers an independently established approximation bound; it derives none, and all information about the true process is carried by the assumed tolerance. Hypothesis (D1) is a property of the triple (proposal, surrogate, temperature) relative to the target, and §7.5 says when it can hold at all. In an application in which the true path law is unknown, (D1) cannot be tested from proposal samples, and the estimator's output is $F^*(\nu)$ and nothing more; §11 discusses what decision consumes $F^*(\nu)$ in that case. An earlier version of this paper claimed an unconditional bound on the selection bias in terms of the integrated deficit of the target family; that claim was withdrawn. When the target is divisible, any such bound would force zero bias for *every* admissible proposal and surrogate, and the selected measure can be made arbitrarily far from a divisible target by choosing a poor proposal or an extreme temperature.

7.3 The obstruction: divisible surrogates cannot match indivisible kernels

Proposition 7.2 (Deficit obstruction to divisible approximation)**.** *In setting (K), let* $\{\Gamma(t \leftarrow s)\}$ *be an indivisible target whose two-time operator family satisfies the standing contraction assumption of [70, Assumption 3.3] on* $L^2(\Omega, \mu)$*, and let* $\{\tilde{\Gamma}(t \leftarrow s)\}$ *be* any *divisible two-time family on* Ω *satisfying the same assumption on the same* $L^2(\Omega, \mu)$*. Then for every triple* $0 \le s < r < t \le T$*,*

$$D(t,s,r) \;\le\; \delta(t,s) + \delta(t,r) + \delta(r,s), \qquad \delta(\tau_1,\tau_2) := \|\Gamma(\tau_1 \leftarrow \tau_2) - \tilde{\Gamma}(\tau_1 \leftarrow \tau_2)\|_{\mathrm{op}}, \tag{22}$$

hence, at every triple with $D(t,s,r) > 0$*,*

$$\sup_{(\tau_1,\tau_2)} \|\Gamma(\tau_1 \leftarrow \tau_2) - \tilde{\Gamma}(\tau_1 \leftarrow \tau_2)\|_{\mathrm{op}} \;\ge\; \frac{D(t,s,r)}{3}.$$

Proof. This is the universal three-pair bound of [70, Corollary 4.6]: insert $\tilde{\Gamma}(t \leftarrow s) = \tilde{\Gamma}(t \leftarrow r)\tilde{\Gamma}(r \leftarrow s)$ into the deficit by the triangle inequality and bound the composite terms using the contraction hypothesis on one factor of each product. □

Remark 7.3 (A sufficient condition for the surrogate). For the family $\tilde{\Gamma}(t \leftarrow s) = e^{(t-s)L_\nu}$ induced by a surrogate pair (b_ν, σ_ν), it suffices that μ be an invariant measure of the surrogate semigroup: then e^{tL_ν} is a contraction on $L^2(\Omega, \mu)$ by Jensen's inequality. For the Ornstein–Uhlenbeck surrogate this is its Gaussian stationary law, and the target's reference measure must then be that same μ.

Proposition 7.2 formalizes Remark 3.4 at the level of the whole family: no divisible surrogate reproduces an indivisible target's kernels uniformly within $D(t,s,r)/3$ at the worst time pair, for any triple with positive deficit. The scope is limited in two respects. First, the statement is a *family-norm* obstruction—the supremum over time pairs is essential; a divisible family may match $\Gamma(t \leftarrow s)$ exactly at one designated pair while paying at the others, so no single-pair lower bound of this form holds, and none is claimed. Second, the bound is an impossibility result about *kernels in operator norm*, in setting (K); it says nothing about the total-variation distance in (D1), which is a statement about path measures in setting (L) and involves the proposal and the temperature as well.

7.4 The obstruction and the selection bias are independent

The July version of this paper opened this discussion with the sentence "the conditional structure of Theorem D is forced, not chosen: no choice of divisible surrogate drives the selection bias to zero, because divisible families are uniformly bounded away from an indivisible target's kernels." That inference is invalid, and we withdraw it. $Q_{\nu,M}$ is not the path law of the surrogate; it is a product

of the proposal law and the tempered surrogate density (Proposition 3.5), and the proposal is a free input. Separation of the surrogate's kernels from the target's, which Proposition 7.2 bounds, and separation of the selected measure from the target's path law, which (D1) bounds, are logically independent quantities. Example 7.4 exhibits an indivisible target with zero selection bias.

Example 7.4 (Zero selection bias with an indivisible target). Take $d = 1$, $M = 8$, $\delta t = 1/8$, the Ornstein–Uhlenbeck surrogate with $k_\nu = 2$, $\sigma_\nu = 1$, $\bar{x} = 0$, and $\lambda = 1/2$. Let the proposal be the centered Gaussian $\Pi_M = \mathcal{N}(0, K_\Pi^{-1})$ on $(\xi_1, \dots, \xi_8)$ with

$$K_\Pi \;=\; \frac{1}{\delta t} A(a_\Pi) \;+\; \varepsilon\big(e_1 e_8^\top + e_8 e_1^\top + e_2 e_7^\top + e_7 e_2^\top\big), \qquad a_\Pi = 1 - \delta t, \quad \varepsilon = 2.5,$$

where $A(a)/\delta t$ is the tridiagonal precision of Corollary 3.6; K_Π has least eigenvalue 0.339, so Π_M is a legitimate, trivially samplable proposal. Let the target be the centered Gaussian process on the grid with precision $K_\Pi + \lambda K_\nu$ (least eigenvalue 1.270). By the product-of-Gaussians step in the proof of Corollary 3.6(i), which uses nothing about the proposal beyond its being centered Gaussian, $Q_{\nu,M} = \mathcal{N}(0, (K_\Pi + \lambda K_\nu)^{-1}) = P_{\mathrm{true},M}$ *exactly*: (D1) holds with $\eta_{\nu,M} = 0$ and the selection bias vanishes identically. The target is nonetheless indivisible. For a centered Gaussian family the two-time kernels compose only if the conditional-mean coefficients do, $C_{ts}/C_{ss} = (C_{tr}/C_{rr})(C_{rs}/C_{ss})$ in terms of the covariance $C = (K_\Pi + \lambda K_\nu)^{-1}$; at the grid triple $(s, r, t) = (3, 5, 7)$ the direct coefficient is -0.008588 against a composed value of 0.305127, an absolute gap of 0.314, and none of the 56 triples composes (the smallest gap is 0.011). The surrogate is divisible, and where the contraction hypothesis of Proposition 7.2 is met its kernels sit at operator-norm distance at least $D/3$ from the target's.

The refutation does not use Proposition 7.2: an indivisible target with zero selection bias refutes any inference from indivisibility to nonzero bias. The construction sets $K_\Pi = K_{\mathrm{target}} - \lambda K_\nu$: zero bias is attained by *proposal* design, and the design presupposes the target's law. The obstruction says that improving the surrogate's kernels cannot close the truth link, and it says nothing more; the proposal can close it only by presupposing the target's law. The design problem of §7.5 concerns the triples between these two cases.

7.5 The design rule, and the open problem in the form it must take

Corollary 3.6 fixes the quadratic variation of the selected measure, and diffusion laws with different quadratic variation are mutually singular; on the grid, Gaussian Markov chains with different innovation variances have total-variation distance tending to one as M grows. Hence:

Proposition 7.5 (Design rule for the transfer hypothesis)**.** *In setting (L), let the target's grid laws $P_{\mathrm{true},M}$ be such that the realized quadratic variation $\sum_{m<M}(\xi_{m+1} - \xi_m)^2$ converges in $P_{\mathrm{true},M}$-probability to $\sigma^2 T$ as $M \to \infty$—as it does for any diffusion with constant diffusion coefficient σ—let the proposal and surrogate be as in Corollary 3.6, and let $M \to \infty$ at fixed $(\lambda, \sigma_\Pi, \sigma_\nu)$. Then $d_{\mathrm{TV}}(Q_{\nu,M}, P_{\mathrm{true},M}) \to 1$ unless*

$$\frac{1}{\sigma_\Pi^2} + \frac{\lambda}{\sigma_\nu^2} \;=\; \frac{1}{\sigma^2}. \tag{23}$$

In particular, with $\sigma_\nu = \sigma$ the proposal must satisfy $\sigma_\Pi^2 = \sigma^2/(1-\lambda)$: it must be rougher than the target by the factor $(1-\lambda)^{-1}$, which diverges as $\lambda \uparrow 1$, and no finite-diffusion proposal serves at $\lambda \geq 1$.

Proof. Write $Y_m = \xi_{m+1} - \xi_m$ $(0 \leq m < M)$ for the increments, so that $\xi_m = \sum_{j<m} Y_j$, and $V_M := \sum_{m<M} Y_m^2$ for the realized quadratic variation. By Proposition 3.5 the density of $Q_{\nu,M}$ in

the coordinates $Y = (Y_0, \dots, Y_{M-1})$ is proportional to

$$\exp\Big\{-\frac{1}{2\delta t}\Big[\frac{1}{\sigma_\Pi^2}\sum_{m<M}\big(Y_m + k_\Pi \delta t\,\xi_m\big)^2 + \frac{\lambda}{\sigma_\nu^2}\sum_{m<M}\big(Y_m + k_\nu \delta t\,\xi_m\big)^2\Big]\Big\}.$$

Expanding the squares and using $2\sum_{m<M} Y_m\xi_m = (\sum_{m<M} Y_m)^2 - \sum_{m<M} Y_m^2$, the exponent is $-\frac{1}{2\delta t}Y^\top G Y$ with

$$G \;=\; u\,I + E, \qquad E \;=\; \delta t\,\kappa\,\mathbf{1}\mathbf{1}^\top + \delta t^2\,\kappa_2\,L^\top L,$$

where $u = a_\Pi/\sigma_\Pi^2 + \lambda a_\nu/\sigma_\nu^2$ is the off-diagonal coefficient in the proof of Corollary 3.6, $\kappa = k_\Pi/\sigma_\Pi^2 + \lambda k_\nu/\sigma_\nu^2$, $\kappa_2 = k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2$, $\mathbf{1}$ is the all-ones vector and L the strictly lower-triangular matrix of ones, so that $(LY)_m = \xi_m$. Since the rates k_Π, k_ν are positive and $k_\bullet \delta t < 1$, $u > 0$ and $E \succeq 0$. Under $Q_{\nu,M}$ the increment vector is therefore centered Gaussian with covariance $C_M = \delta t\, G^{-1} \preceq (\delta t/u)\, I$, and

$$\mathbb{E}_{Q_{\nu,M}}[V_M] = \operatorname{tr} C_M, \qquad \operatorname{Var}_{Q_{\nu,M}}(V_M) = 2\operatorname{tr}(C_M^2) \;\le\; 2M\|C_M\|_{\mathrm{op}}^2 \;\le\; \frac{2T\delta t}{u^2}.$$

For the mean, $G^{-1} = u^{-1}(I - EG^{-1})$ gives $\operatorname{tr} G^{-1} = M/u - u^{-1}\operatorname{tr}(E^{1/2}G^{-1}E^{1/2})$ with $0 \le \operatorname{tr}(E^{1/2}G^{-1}E^{1/2}) \le \|G^{-1}\|_{\mathrm{op}}\operatorname{tr} E \le \operatorname{tr} E/u$, and $\operatorname{tr} E = \kappa T + \kappa_2 T(T-\delta t)/2$; so

$$\frac{T}{u} - \frac{\delta t}{u^2}\Big(\kappa T + \frac{\kappa_2 T^2}{2}\Big) \;\le\; \mathbb{E}_{Q_{\nu,M}}[V_M] \;\le\; \frac{T}{u}.$$

As $M \to \infty$, $u \to 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 =: 1/\sigma_\infty^2$, the limit of $1/\sigma_{\mathrm{eff}}^2$ in (10); so $\mathbb{E}_{Q_{\nu,M}}[V_M] \to \sigma_\infty^2 T$ and $\operatorname{Var}_{Q_{\nu,M}}(V_M) \to 0$, and $V_M \to \sigma_\infty^2 T$ in $Q_{\nu,M}$-probability by Chebyshev's inequality. If $\sigma_\infty^2 \neq \sigma^2$, the event $\{|V_M - \sigma_\infty^2 T| < \varepsilon\}$ with $\varepsilon < |\sigma_\infty^2 - \sigma^2|T$ has $Q_{\nu,M}$-probability tending to one and $P_{\mathrm{true},M}$-probability tending to zero, so $d_{\mathrm{TV}} \to 1$. The rest is (10). (If the target's realized quadratic variation converges in $P_{\mathrm{true},M}$-probability to a random limit V, the same argument gives $\liminf_M d_{\mathrm{TV}}(Q_{\nu,M}, P_{\mathrm{true},M}) \ge 1 - \mathbb{P}(V = \sigma_\infty^2 T)$, so $d_{\mathrm{TV}} \to 1$ whenever the law of V has no atom at $\sigma_\infty^2 T$.) □

The natural proposals of this paper—forward simulators, neural-SDE generators, bridge constructions built to produce plausible candidate paths—share the *target's* roughness, not the roughness (23) prescribes. For Gaussian pairs Proposition 7.5 shows that the transfer hypothesis then fails under refinement whatever the surrogate's rate; the proposition is stated and proved for Gaussian pairs only, and is a necessary condition, not a characterization. The same mechanism is expected for general proposals and surrogates with constant diffusion coefficients—the normalized selected chain is a Doob transform of the one-step product kernel, which changes drift but not diffusion—but that extension is not proved here. In the Gaussian example, where no target is specified, the total-variation distance between the selected measure and the proposal itself is bounded below through the Bhattacharyya coefficient $\varrho = \int \sqrt{dQ\, d\Pi}$, $d_{\mathrm{TV}} \ge 1 - \varrho$ [72, Lemma 2.3], with ϱ in closed form for centered Gaussians, $\varrho = \det(\Sigma_Q)^{1/4}\det(\Sigma_\Pi)^{1/4}\det(\frac{1}{2}(\Sigma_Q + \Sigma_\Pi))^{-1/2}$:

M	25	50	100	200	400
ϱ	4.73×10^{-1}	2.27×10^{-1}	5.20×10^{-2}	2.73×10^{-3}	7.57×10^{-6}
$d_{\mathrm{TV}}(Q_{\nu,M}, \Pi_M) \;\ge$	0.527	0.774	0.948	0.9973	0.99999

So for the target equal in law to the proposal itself—the case of a perfect simulator—the tolerance in (D1) exceeds 0.77 already at $M = 50$, and the bound $2F_{\max}\eta_{\nu,M}$ exceeds the range of the observable; for any other target sharing the proposal's quadratic variation, Proposition 7.5 gives the same conclusion under refinement, while at a fixed grid the two total-variation distances are

related only by the triangle inequality. This does not contradict Theorem 7.1, which is a correct conditional statement, and it does not apply at a fixed coarse grid; it fixes the form of the open problem.

Remark 7.6 (The open bridge, restated). The July version asked whether the tolerance $\eta_{\nu,M}$ can be controlled by indivisibility invariants of the target—the integrated deficit $J(\mathcal{R}) = \iiint_{s<r<t} D(t,s,r)^2 \, ds\, dr\, dt$ of [70, Theorem 7.4]—"for principled choices of proposal and surrogate." Stated so, the question is not well posed: no invariant of the target can control a quantity that depends on the proposal–surrogate–temperature triple, since a poor proposal or an extreme temperature makes $Q_{\nu,M}$ arbitrarily far from any target, divisible or not, and Proposition 7.5 shows that the class of admissible triples is constrained by the target's quadratic variation before any invariant enters. The problem that remains is a *design* problem at fixed M: for the admissible class of triples whose selected quadratic variation matches the target's, whether the residual tolerance is controlled by the target's indivisibility invariants *together with* separately measured proposal and surrogate errors. Stated so, it is open. The present paper supplies the estimator-side theory (Theorems A–C), the constraint on the class (Proposition 7.5), and the precise statement of what remains; no result above depends on it.

8 Surrogate characterizations and calibration

We characterize three surrogate families and formulate the calibration of ν from data.

8.1 Ornstein–Uhlenbeck surrogate

Take $b_\nu(x) = -k(x - \bar{x})$ with $\nu = (k, \bar{x}, \sigma) \in (0,\infty) \times \mathbb{R}^d \times \mathrm{Sym}_d^+$ and $\sigma_\nu = \sigma$ constant. The divergence is constant, $\nabla \cdot b_\nu = -kd$, so Lemma 3.9 applies with $W_{\max} = e^{\lambda k T d/2}$ and the selected measure exists with finite chi-square for every proposal, and Proposition 3.5 applies in its constant-divergence form: the selection weight is the tempered Euler path density of the surrogate, and the divergence term cancels identically from the estimator. The discrete action (7) is

$$S_{\nu,M}[\xi] \;=\; \frac{1}{2} \sum_{m=0}^{M-1} \frac{\left|\sigma^{-1}\big(\xi_{m+1} - \xi_m + k(\xi_m - \bar{x})\,\delta t\big)\right|^2}{\delta t} \;-\; \frac{kTd}{2}, \tag{24}$$

a quadratic functional of ξ; when the proposal is itself Gaussian on the grid, the selected measure is the Gaussian Markov chain of Corollary 3.6. The restoring rate k is the surrogate's single tunable timescale. The action accumulates over the entire horizon $[0, T]$, with no window truncation in (24), and k controls an effective decay of *sensitivity*: the influence of an increment on the surrogate's implied conditional mean decays on the timescale $1/k$. The functional itself is not local in time, consistent with Proposition 3.5: a path density is not a local object.

8.2 Polynomial-drift surrogate

Take $b_\nu(x) = \sum_{|\alpha| \le M_p} c_\alpha \, x^\alpha$ for multi-indices $\alpha \in \mathbb{N}_0^d$ of degree at most M_p, and $\sigma_\nu = \sigma$. For $M_p = 1$ with $b_\nu(x) = Ax + a$, $A \prec 0$, this recovers the multivariate OU surrogate with restoring matrix $-A \succ 0$ and $\bar{x} = -A^{-1}a$; $M_p \ge 2$ admits double-well and metastable structure. For $M_p \ge 2$ the divergence $\nabla \cdot b_\nu$ is in general a non-constant polynomial and can be unbounded below—as for the double-well drift $b(x) = x - x^3$ on $\mathbb{R}$, with $\nabla \cdot b = 1 - 3x^2$. In that case Lemma 3.9's automatic bound is unavailable and Assumption 3.2(P2) must be certified for the proposal at hand. For the class of drifts whose divergence is bounded below by a quadratic—which contains every polynomial

drift of degree at most three, since the divergence is then a polynomial of degree at most two—and for Gaussian proposals, the following criterion certifies (P2).

Lemma 8.1 ((P2) certificate for quadratically bounded divergence under Gaussian proposals)**.** *Suppose* $(\nabla\cdot b_\nu)(x) \geq -c_2|x|^2 - c_0$ *for all* $x \in \Omega$*, and suppose that under* Π_M *the vector* $(\xi_1, \ldots, \xi_M)$ *is Gaussian with covariance* Σ_Π *(any mean), with* ξ_0 *deterministic. Then* $\mathbb{E}_{\Pi_M}[w_\nu^2] < \infty$*, hence (P2) holds, whenever*

$$\lambda\, c_2\, \delta t\, \|\Sigma_\Pi\|_{\mathrm{op}} \;<\; \tfrac{1}{2}. \tag{25}$$

Proof. Since the kinetic sum is non-negative, $w_\nu^2 = e^{-2\lambda S^{\mathrm{kin}}_{\nu,M} - 2\lambda S^{\mathrm{div}}_{\nu,M}} \leq e^{-2\lambda S^{\mathrm{div}}_{\nu,M}} \leq \exp\big(\lambda c_0 T + \lambda c_2 \delta t \sum_{m<M} |\xi_m|^2\big) \leq e^{\lambda c_0 T + \lambda c_2 \delta t |\xi_0|^2} \exp(\lambda c_2 \delta t\, |\xi|^2)$ with $\xi = (\xi_1, \ldots, \xi_M)$. For a Gaussian vector with covariance Σ, $\mathbb{E}\exp(s|\xi|^2) < \infty$ if and only if $I - 2s\Sigma \succ 0$, i.e. $s < 1/(2\|\Sigma\|_{\mathrm{op}})$; apply this with $s = \lambda c_2 \delta t$. □

The certificate discards the kinetic sum entirely and is correspondingly crude; it is a sufficient condition, and its failure settles nothing. In the double-well example of §9.5, $c_2 = 3$, $c_0 = 0$, $\delta t = 0.02$ and $\|\Sigma_\Pi\|_{\mathrm{op}} = 10.04$, so (25) reads $0.602\lambda < 0.5$: it certifies (P2) at $\lambda = 3/4$, the temperature used there, and fails at $\lambda = 1$, where the question is left open.

8.3 Mori–Zwanzig surrogate by explicit Markovian closure

A more principled surrogate derives from the Mori–Zwanzig (MZ) formalism [56, 81, 82, 83, 17, 32, 68, 51, 52]. Let P be a projection onto a space of resolved observables u. The MZ identity writes the resolved evolution as

$$\dot u(t) \;=\; \Phi(u(t)) \;-\; \int_0^t \kappa(t-s)\, u(s)\, ds \;+\; f(t),$$

a Markovian term Φ generated by the projected dynamics, a memory integral against a kernel κ supported on the past, and a fluctuating force f orthogonal to the resolved subspace, with, in the Mori (linear) projection, the second fluctuation–dissipation relation $\mathbb{E}[f(t)f(s)^\top] = \kappa(t-s)\, C$ for $t \geq s$ and the resolved stationary covariance C [83]. The formalism says that projection produces memory and does not remove it. The memory integral comes from eliminating degrees of freedom, the projected dynamics is generically non-Markovian, and no divisible kernel arises from projection alone. A divisible surrogate is obtained only by a *deliberate closure*, and the closure consists of two steps. (i) The memory integral is replaced by its Markovian approximation: the kernel is collapsed to a delta function with the same integral, $\gamma := \int_0^\infty \kappa(\tau)\, d\tau$, so the memory term becomes $-\gamma\, u(t)$. (ii) The fluctuating force is *modeled*, not dropped: it is replaced by white noise whose intensity is the zero-frequency spectral density of the force, $\sigma\sigma^\top = \int_{\mathbb{R}} R_f(t)\, dt = \int_0^\infty \big(R_f(t) + R_f(t)^\top\big)\, dt$ with $R_f(t) = \mathbb{E}[f(t)f(0)^\top]$ and $R_f(-t) = R_f(t)^\top$, which the fluctuation–dissipation relation evaluates in the Mori case to $\gamma C + C\gamma^\top$; the one-sided integral $\int_0^\infty R_f(t)\, dt = \gamma C$ alone is neither symmetric nor positive semidefinite in general. The surrogate pair is then

$$b_{\nu^P}(u) \;=\; \Phi(u) - \gamma u, \qquad \sigma_{\nu^P}\sigma_{\nu^P}^\top \;=\; \gamma C + C\gamma^\top,$$

and $\sigma\sigma^\top$ is positive semidefinite by Bochner's theorem, being a spectral density; the non-degeneracy $\sigma\sigma^\top \succ 0$ required by Definition 3.3 is an assumption on the projection, stated as such (it is not implied by $C \succ 0$ and a dissipative γ alone). For linear Φ the relation $\gamma C + C\gamma^\top = \sigma\sigma^\top$ is the Lyapunov equation of the resulting Ornstein–Uhlenbeck process; for nonlinear Φ it is the fluctuation–dissipation-consistent white-noise intensity and nothing more. Dropping the fluctuating

force instead of modeling it leaves the deterministic term $\Phi(u) - \gamma u$ with $\sigma = 0$, which is not a surrogate in the sense of Definition 3.3; the July version of this paper described the closure that way and was wrong to do so. Both steps are additional approximations and are stated as such: (i) discards the memory carried by the kernel, which is the source of the Chapman–Kolmogorov failure at the two-time-family level, and (ii) replaces a colored force by a white one. Proposition 7.2 bounds how well *any* such divisible closure—MZ-derived or otherwise—can track an indivisible target's kernels: uniformly in operator norm, at the worst time pair, by $D(t, s, r)/3$. The obstruction says nothing more about the closure: nothing about other metrics, particular observables, restricted time pairs, or the regime in which the target's deficit tends to zero. The MZ surrogate is, as Remark 3.4 says of every surrogate, a reweighting object and not an approximation scheme that converges uniformly to the truth.

8.4 Calibration as quasi-likelihood M-estimation

When observed sample paths $\xi^{(1)}_{\mathrm{obs}}, \dots, \xi^{(n)}_{\mathrm{obs}}$ are available—in setting (L), data from the inducing process; in applications, the empirical record on the grid—the surrogate parameter ν may be calibrated from them. The July version of this paper proposed minimizing the empirical surrogate action $\frac{1}{n}\sum_i S_{\nu,M}[\xi^{(i)}_{\mathrm{obs}}]$ and called its population minimizer ν^* "the best reweighting parameter within the chosen family." That proposal is withdrawn for two reasons.

Inconsistency. Even when the observed law lies in the surrogate family, the population minimizer of the expected action is not the true parameter. In the Gaussian example of §9, take the observed law to be the OU proposal itself ($k_\Pi = 1$, $\sigma_\Pi = 1$) and sweep the surrogate's restoring rate k at $\sigma_\nu = 1$. Since $\xi_{m+1} - (1 - k\delta t)\xi_m = (k-1)\delta t\, \xi_m + \sqrt{\delta t}\, \eta_m$ under the proposal,

$$\mathbb{E}_{\Pi_M}\big[S_{k,M}\big] \;=\; (k-1)^2\, \frac{\delta t}{2} \sum_{m=0}^{M-1} \mathrm{Var}_{\Pi_M}(\xi_m) \;+\; \frac{M}{2} \;-\; \frac{kT}{2},$$

minimized at $k^* = 1 + \big[2\delta t \sum_m \mathrm{Var}_{\Pi_M}(\xi_m)\big]^{-1} = 2.76$, not at the true rate 1: the OM divergence term $-kT/2$ tilts the objective. The kinetic sum alone is minimized at $k = 1.000$. *Ill-posedness in the diffusion coefficient.* Under $\sigma_\nu = cI$ the kinetic sum scales as c^{-2} while the divergence sum is independent of c, so minimizing $S_{\nu,M}$ over c drives c to the boundary of the compact parameter set. Both defects have one cause: by Proposition 3.5, $S_{\nu,M}$ is a negative log-density only up to a normalization it lacks and a divergence term it carries, and minimizing it is not M-estimation of any quantity one wants.

Calibration is therefore formulated through the surrogate's own Euler quasi-likelihood. With p_ν as in Proposition 3.5, set $\ell_{\nu,M}[\xi] := -\log p_\nu(\xi) = S^{\mathrm{kin}}_{\nu,M}[\xi] + \frac{M}{2}\log\det(2\pi\delta t\, \sigma_\nu \sigma_\nu^\top)$ and

$$\hat{\nu}_{\mathrm{QML}} \;=\; \arg\min_{\nu \in \Theta_0} \; \frac{1}{n} \sum_{i=1}^{n} \ell_{\nu,M}\big[\xi^{(i)}_{\mathrm{obs}}\big], \tag{26}$$

the quasi-maximum-likelihood estimator for discretely observed diffusions [28, 45], with the misspecified case treated in [73]. The normalization term is increasing in σ_ν and the kinetic term decreasing, so the objective has an interior minimizer in the diffusion coefficient; when the observed law is the surrogate's Euler chain the estimator is consistent for the true parameter ($k = 1.000$ in the sweep above). When it is not, its population limit $\nu_{\mathrm{QML}} = \arg\min_\nu D_{\mathrm{KL}}(P_{\mathrm{obs},M} \,\|\, p_\nu)$ is the Kullback–Leibler projection of the observed grid law onto the surrogate family—a misspecified-model projection in White's sense [79], which requires only that the observed law exist and be

absolutely continuous with respect to Lebesgue measure on Ω^M, and not that it be generator-based. The July version denied ν^* any interpretation as "the Kullback–Leibler projection of a true generator-based law onto the surrogate family, because no such law exists on the observed description"; the denial was wrong: a KL projection needs a true probability measure, which setting (L) supplies, and not a generator-based one. The classical consistency and sandwich-covariance theorems still do not apply automatically; they require (a) observations that are i.i.d. or satisfy suitable mixing conditions—observed paths may be serially dependent, and a single long realization places the entire burden on the mixing hypotheses; (b) identification, i.e. a unique population minimizer on the compact Θ_0; (c) domination and twice continuous differentiability of $\nu \mapsto \ell_{\nu,M}$ near it; (d) a non-singular Hessian there; and (e) a uniform law of large numbers over Θ_0. Under (a)–(e), $\hat{\nu}_{\mathrm{QML}} \to \nu_{\mathrm{QML}}$ and $\sqrt{n}(\hat{\nu}_{\mathrm{QML}} - \nu_{\mathrm{QML}})$ is asymptotically normal with sandwich covariance [43, 79, 58].

Calibration and selection are different operations, and the separation is deliberate. Calibration fixes ν from data through the normalized quasi-likelihood; selection then reweights candidate paths by the OM action $S_{\nu,M}$ at that ν, divergence term included. Three parameters are thereby distinguished, and the word "optimal" is reserved for the second:

- ν_{QML}, the *calibrated* parameter of (26), with no claim regarding the estimator's variance;
- $\nu^\dagger := \arg\min_\nu \sigma^2(\nu)$ for a fixed observable F, the *variance-optimal* surrogate, an unconditional variance-only optimization for which Theorem C supplies the governing quantities; for a class $\mathcal{F}$ of observables an objective over the class must be specified, $\arg\min_\nu \sup_{F\in\mathcal{F}} \sigma_F^2(\nu)$ or an averaged criterion, and different objectives give different optima;
- ν^*, the withdrawn action-risk minimizer.

In the Gaussian example these are far apart: $k_{\mathrm{QML}} = 1.000$; $k^* = 2.76$ with $\sigma^2 = 20.3$; $k^\dagger = 12.39$ with $\sigma^2 = 4.999$ for $F = \xi_M^2 - 1$; and the $k_\nu = 2$ of §9 has $\sigma^2 = 30.89$. The calibrated and the variance-optimal surrogates answer different questions, and neither is the other.

9 Two worked examples

We exhibit the full machinery on two one-dimensional examples in which the benchmark is exact. In the first, the proposal and the surrogate are both Ornstein–Uhlenbeck, every quantity is in closed form, and the identity of Corollary 3.6 is displayed numerically. In the second, the surrogate is a double well: the drift divergence is non-constant, the OM divergence term is exercised, and the benchmark is obtained by transfer-matrix quadrature. Both examples use the same proposal, grid and observable. In both, the proposal's transition density is available in closed form; the estimator does not use it, but the benchmarks do.

9.1 Example I: setup

Set $\Omega = \mathbb{R}$, $T = 1$, and fix the grid $t_m = m\,\delta t$ with $M = 50$ and $\delta t = T/M = 0.02$. The proposal is the Euler–Maruyama discretization of the OU process with restoring rate $k_\Pi = 1$ and noise $\sigma_\Pi = 1$, *started deterministically at* $\xi_0 = 0$:

$$\xi_{m+1} \;=\; a_\Pi\,\xi_m + \sigma_\Pi\sqrt{\delta t}\,\eta_m, \qquad a_\Pi = 1 - k_\Pi\,\delta t, \qquad \eta_m \sim \mathcal{N}(0,1) \text{ i.i.d.}$$

The start is deterministic and the proposal is not stationary: under Π_M the terminal variance is

$$\mathrm{Var}_{\Pi_M}(\xi_M) \;=\; \delta t\,\frac{1 - a_\Pi^{2M}}{1 - a_\Pi^2} \;=\; 0.438071,$$

the discrete analogue of the continuous-time value $(1-e^{-2})/2 \approx 0.432332$ for an OU process released from the origin, and well below the stationary variance $\frac{1}{2}$. The surrogate is OU with $k_\nu = 2$, $\sigma_\nu = 1$, $\bar{x} = 0$; the temperature is $\lambda = 1$. With $\nabla \cdot b_\nu = -k_\nu$ and $a_\nu = 1 - k_\nu\,\delta t$, the discrete action and weight (24) are

$$S_{\nu,M}[\xi] \;=\; \frac{1}{2\,\delta t}\sum_{m=0}^{M-1}\big(\xi_{m+1} - a_\nu\,\xi_m\big)^2 \;-\; \frac{k_\nu T}{2}, \qquad w_\nu(\xi) \;=\; e^{-\lambda S_{\nu,M}[\xi]} \;\le\; W_{\max} = e^{\lambda k_\nu T/2} = e.$$

The observable is the quadratic terminal functional $F(\xi) = \xi_M^2 - 1$; it is unbounded, with all moments finite under the Gaussian measures below, so Theorem A and the probability bound of Theorem B–I apply exactly, while the bounded-observable statements apply to truncations $F \wedge R$.

9.2 Example I: closed forms, and the identity displayed

Conditionally on $\xi_0 = 0$, both Π_M and the weight are Gaussian in $(\xi_1, \dots, \xi_M) \in \mathbb{R}^M$: $\Pi_M = \mathcal{N}(0, K_\Pi^{-1})$ and $w_\nu \propto \exp(-\frac{\lambda}{2\delta t}\sum_m(\xi_{m+1} - a_\nu\xi_m)^2)$, with K_Π and K_ν the tridiagonal precision matrices of the two quadratic forms. Hence the selected measure and the w^2-tilted measure are themselves Gaussian:

$$Q_{\nu,M} \;=\; \mathcal{N}\big(0,\ \Sigma_\lambda\big), \quad \Sigma_\lambda = (K_\Pi + \lambda K_\nu)^{-1}; \qquad \frac{w_\nu^2\, d\Pi_M}{\mathbb{E}[w_\nu^2]} \;=\; \mathcal{N}\big(0,\ \Sigma_{2\lambda}\big), \quad \Sigma_{2\lambda} = (K_\Pi + 2\lambda K_\nu)^{-1}.$$

This is Corollary 3.6 in concrete form: $K_\Pi + \lambda K_\nu$ is tridiagonal, so $Q_{\nu,M}$ is a Gaussian Markov chain, and its interior rows are those of an OU Euler chain with $\sigma^2_{\text{eff}} = 0.4992$ (the leading-order value $(1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2)^{-1} = 0.5$) and $k_{\text{eff}} = 1.58$ (from $k^2_{\text{eff}}/\sigma^2_{\text{eff}} = k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2 = 5$ exactly). The increment variance of a bulk step is 0.0101 under $Q_{\nu,M}$ against 0.0201 under Π_M: the selected measure is a smoother process than the proposal, with half its quadratic variation. Three exact quantities follow, none requiring any sampling.

The estimand. With $c := (\Sigma_\lambda)_{MM} = 0.158061$ the selected terminal variance,

$$F^*(\nu) \;=\; \mathbb{E}_{Q_{\nu,M}}[\xi_M^2] - 1 \;=\; c - 1 \;=\; -0.841939. \tag{27}$$

This exact evaluation is the benchmark for the convergence study below; no Monte Carlo benchmark is used. For comparison, the proposal's own value is $\mathbb{E}_{\Pi_M}[\xi_M^2] - 1 = -0.561929$; the selection displaces the estimand by 0.280.

The chi-square divergence. The divergence-correction constants cancel in the ratio and

$$\chi^2\big(Q_{\nu,M}\,\|\,\Pi_M\big) \;=\; \frac{\det(K_\Pi + \lambda K_\nu)}{\det(K_\Pi)^{1/2}\,\det(K_\Pi + 2\lambda K_\nu)^{1/2}} \;-\; 1 \;=\; 1372.4, \tag{28}$$

with the Gaussian Kullback–Leibler divergence likewise available in closed form: $D_{\text{KL}}(Q_{\nu,M}\|\Pi_M) = 4.860$, so $\exp(D_{\text{KL}}) \approx 129.0$, consistent with Theorem 6.1(iii).

The asymptotic variance. By Theorem 6.1(i) and Gaussian fourth moments, with $s^2 := (\Sigma_{2\lambda})_{MM} = 0.096803$,

$$\sigma^2(\nu) \;=\; \frac{\mathbb{E}_{\Pi_M}\big[(F - F^*)^2 w_\nu^2\big]}{Z_\nu^2} \;=\; (1+\chi^2)\,\mathbb{E}_{\mathcal{N}(0,s^2)}\big[(\xi^2 - c)^2\big] \;=\; (1+\chi^2)\big(3s^4 - 2c\,s^2 + c^2\big) \;=\; 30.89, \tag{29}$$

so the CLT predictor is $\sqrt{\sigma^2(\nu)} \approx 5.56$. Formula (29) integrates the squared deviation against the w^2-tilted measure, not the selected measure: the incorrect factorization $(1 + \chi^2)\,\mathrm{Var}_{Q_{\nu,M}}(F)$

(Remark 4.3) would give $(1+\chi^2)\cdot 2c^2 = 68.6$ here, more than twice the true value. Finally $Z_\nu = 7.87\times10^{-8}$, so $W_{\max}/Z_\nu \approx 3.5\times10^7$: the weights are bounded (Tier II applies in principle) but the bounded-weight exponents of (19)–(20) are vacuous at practical N, and the chi-square-based Tier I and the diagnostics of Theorem C carry the operational content, exactly as Remark 5.4 anticipates.

9.3 Example I: numerical convergence study

We compute $\hat{F}_N(\nu)$ for $N = 10^2, 3\times10^2, 10^3, 3\times10^3, 10^4, 3\times10^4, 10^5$ over 1,000 independent replications and compare against the exact benchmark (27). Table 1 reports the absolute empirical bias, the standard deviation, the $\sqrt{N}$-scaled standard deviation with a bootstrap standard error, the range of that statistic over the twenty disjoint batches of 50 replications contained in the 1,000, and the median effective sample size $\hat{N}_{\text{eff}}$. The batch-of-50 column explains why the replication count was raised from 50: at fifty replications the $\sqrt{N}\cdot$ std statistic varies by a factor of two to three from batch to batch, and one batch cannot support any statement about its approach to the asymptote.

N	|bias|	std($\hat{F}_N$)	$\sqrt{N}\cdot$ std ($\pm$ s.e.)	batch-of-50 range	median $\hat{N}_{\text{eff}}$
100	4.68×10^{-2}	1.57×10^{-1}	1.57 ± 0.07	[1.09, 2.25]	3.1
300	3.00×10^{-2}	1.22×10^{-1}	2.12 ± 0.13	[1.53, 3.14]	4.8
1 000	1.97×10^{-2}	9.13×10^{-2}	2.89 ± 0.14	[2.09, 3.90]	7.8
3 000	1.06×10^{-2}	6.34×10^{-2}	3.47 ± 0.24	[2.42, 7.08]	13.6
10 000	4.68×10^{-3}	3.95×10^{-2}	3.95 ± 0.14	[2.66, 4.75]	28.2
30 000	2.79×10^{-3}	2.93×10^{-2}	5.07 ± 0.48	[3.30, 10.45]	57.7
100 000	4.36×10^{-4}	1.67×10^{-2}	5.29 ± 0.17	[3.51, 6.24]	142.6

Table 1: Example I: convergence of $\hat{F}_N(\nu)$ over 1,000 replications (seed 20260914), against the *exact* benchmark $F^*(\nu) = -0.841939$ of (27). The $\sqrt{N}\cdot$ std column rises toward the exact CLT asymptote $\sqrt{\sigma^2(\nu)} \approx 5.56$ of (29) and is 5% below it at $N = 10^5$, with a bootstrap standard error of 0.17: the empirical variance is dominated by rare large weights and is negatively biased at finite N, and the bias is resolved only slowly. The batch-of-50 column shows why fifty replications cannot support that statement. Median $\hat{N}_{\text{eff}}/N \approx 1.4\times10^{-3}$ at $N = 10^5$, above its almost-sure limit $1/(1+\chi^2) = 7.3\times10^{-4}$ (Theorem 6.1(ii)) for the same reason.

9.4 Example I: effective sample size, the two scales, and refinement

Evaluating the closed forms exactly gives

$$\chi^2\big(Q_{\nu,M}\,\|\,\Pi_M\big) \;=\; 1372.4, \qquad D_{\mathrm{KL}}\big(Q_{\nu,M}\,\|\,\Pi_M\big) \;=\; 4.860.$$

The Chatterjee–Diaconis scale is $\exp(D_{\mathrm{KL}}) \approx 129$; the Sanz-Alonso scale is $1+\chi^2 \approx 1373$; the example lands in the chi-square-binding regime anticipated in §6, the two scales differing by an order of magnitude. A plug-in Monte Carlo estimate of (28) from a single $N = 2\times10^5$ proposal sample reads $\hat{\chi}^2 \approx 1059$ (a second, independent draw in the July version read 623): the plug-in estimator is itself dominated by the heavy upper tail of w_ν, severely negatively biased at this sample size, and highly variable from draw to draw, and the median $\hat{N}_{\text{eff}}/N \approx 1.4\times10^{-3}$ of Table 1 is the same biased diagnostic (its almost-sure limit is $1/(1+\chi^2) = 7.3\times10^{-4}$). The chi-square diagnostic that sets the binding sample-size scale cannot be read reliably off a moderate proposal sample in the

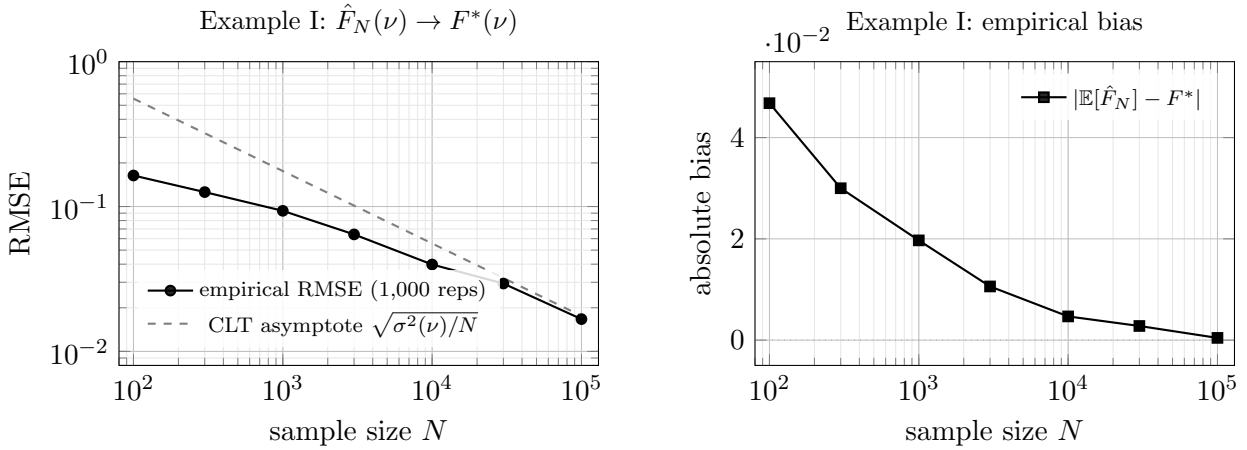


Figure 1: Example I, 1,000 replications, against the exact benchmark $F^*(\nu)$ of (27). Left: empirical RMSE versus N (log–log) with the exact CLT reference $\sqrt{\sigma^2(\nu)/N}$ from (29); the empirical curve approaches the asymptote from below as the weight tail is progressively resolved. Right: absolute empirical bias versus N (log–linear). The figure is a numerical illustration consistent with Theorem 4.1; it does not verify an almost-sure limit theorem.

heavy-tailed regime for which the reweighting object is needed, so the closed form (28) is necessary. The empirical convergence in Table 1 is consistent with the exact scales: at $N = 10^2 \approx \exp(D_{\mathrm{KL}})$ the estimator carries std ≈ 0.16, while stabilization to std $\lesssim 0.04$ occurs for $N \gtrsim 10^4 \approx 7 \times (1+\chi^2)$.

Grid refinement: the mechanism With $\lambda = 1$ and the physical parameters held fixed, refining the grid gives:

M	25	50	100	200	400	800
$1+\chi^2$ (product weight)	3.77×10^{1}	1.37×10^{3}	1.82×10^{6}	3.22×10^{12}	1.00×10^{25}	9.77×10^{49}
Z_ν	4.6×10^{-4}	7.9×10^{-8}	2.3×10^{-15}	2.1×10^{-30}	1.6×10^{-60}	1.0×10^{-120}
χ^2 (ratio weight p_ν/p_Π)	0.1645	0.1601	0.1580	0.1569	0.1564	0.1562

The normalizing constant collapses and the chi-square divergence of the selection weight explodes, exponentially in M, at rate $1 + \chi^2 \approx e^{0.144\,M}$ for these parameters—a rate that depends on the proposal, the surrogate, the temperature and the discretization, and is not a universal constant. The third row is the control: the Girsanov ratio of the surrogate's and the proposal's path densities on the same grids has bounded chi-square divergence converging to a finite continuum value. The mechanism is Corollary 3.6, not Proposition 2.3: the ratio weight leaves the quadratic variation of the proposal untouched, while the product weight halves it, and two Gaussian chains whose innovation variances differ by a fixed factor separate exponentially fast in M. The July version attributed the blow-up to the continuum degeneracy of the pointwise weight; that attribution is replaced. At any fixed M every quantity is finite and Theorems A–C apply verbatim. The selected measures converge: the estimand $F^*(\nu)$ moves from -0.83903 at $M = 25$ through -0.84194 at $M = 50$ to -0.84468 at $M = 1600$, with $O(\delta t)$ error toward the limit -0.84476 (Richardson-extrapolated from $M \le 6400$; it is the terminal-variance readout of the Ornstein–Uhlenbeck law with $\sigma^2_{\mathrm{eff}} = 1/2$ and $k_{\mathrm{eff}} = \sqrt{5/2}$ released from the origin, whose terminal variance 0.1514 is tilted to 0.1552 by the endpoint factor $\exp(0.081\,\xi_T^2)$ that the last row of Corollary 3.6 produces). The limit is singular to the proposal's, so no importance-sampling representation of it from Π_M survives at any fixed $\lambda > 0$; §11 records that a temperature schedule $\lambda_M \downarrow 0$ does not supply one.

9.5 Example II: a double-well surrogate, exercising the divergence term

Keep the proposal, grid and observable of Example I and replace the surrogate by the double-well drift $b_\nu(x) = x - x^3$ with $\sigma_\nu = 1$, whose divergence $\nabla \cdot b_\nu = 1 - 3x^2$ is non-constant and unbounded below; take $\lambda = 3/4$. By Proposition 3.5 the divergence sum $S^{\mathrm{div}}_{\nu,M}$ is now a functional of the path, and the selection action differs from the tempered Euler log-density by a path-dependent term; Example I cannot exercise this term because there it is a constant. Lemma 3.9 does not apply, but Lemma 8.1 does: $c_2 = 3$, $\|\Sigma_\Pi\|_{\mathrm{op}} = 10.04$ and $\lambda c_2 \delta t \|\Sigma_\Pi\|_{\mathrm{op}} = 0.452 < 0.5$, so (P2) holds and Theorems A, B–I and C apply. The certificate fails at $\lambda = 1$ (0.602), so $\lambda = 3/4$ is used.

Benchmark by transfer-matrix quadrature No closed form is available, but none is needed for a deterministic benchmark. The unnormalized selected density on the grid factorizes as a product of one-step kernels $\pi(x' \mid x)\, e^{-\lambda s(x,x')}$ with π the proposal's Gaussian transition density and $s(x,x') = (x' - x - b_\nu(x)\delta t)^2/(2\delta t) + \frac{1}{2}(1 - 3x^2)\delta t$, so Z_ν, $\mathbb{E}_{\Pi_M}[w_\nu^2]$, $\mathbb{E}_{Q_{\nu,M}}[\xi_M^2]$, $\mathbb{E}_{Q_{\nu,M}}[S_{\nu,M}]$ and $\mathbb{E}_{\Pi_M}[(F - F^*)^2 w_\nu^2]$ are all obtained by propagating the appropriate densities forward through $M = 50$ one-step integral operators, discretized on a mesh of 1,801 nodes on $[-4.5, 4.5]$ by the trapezoidal rule. The result is stable to six significant figures under refinement to 2,401 nodes on $[-5, 5]$. The benchmark quantities are

$$Z_\nu = 6.915 \times 10^{-7}, \qquad F^*(\nu) = -0.547760, \qquad \chi^2 = 162.1,$$

$$D_{\mathrm{KL}} = 3.294 \;\; (\exp(D_{\mathrm{KL}}) = 27.0), \qquad \sigma^2(\nu) = 44.31.$$

Deleting the divergence sum from the action—using the tempered Euler log-density alone—gives instead $F^* = -0.637163$, $\chi^2 = 169.3$ and $\sigma^2 = 26.60$. The divergence term moves the estimand by 0.089—more than the entire displacement of either variant from the proposal's value -0.561929—and it reverses the sign of that displacement: with the OM term the selection raises the selected terminal variance relative to the proposal's (from 0.438 to 0.452), while the tempered Euler likelihood alone lowers it (to 0.363). In this sense the construction is an Onsager–Machlup selection rather than a tempered Euler likelihood; Example I could not show it because there the term is a constant.

Convergence study Table 2 reports the same statistics as Table 1, over 1,000 replications, against the quadrature benchmark. The chi-square divergence is an order of magnitude smaller than in Example I, because $\lambda = 3/4$ moves the selected quadratic variation less far from the proposal's (the leading-order formula of Corollary 3.6 with $\sigma_\nu = 1$ gives $\sigma^2_{\mathrm{eff}} \approx 0.57$ against 0.5; the corollary itself is for Gaussian pairs), and the effective sample sizes are correspondingly larger: median $\hat{N}_{\mathrm{eff}}/N \approx 7.3 \times 10^{-3}$ at $N = 10^5$ against the limit $1/(1 + \chi^2) = 6.1 \times 10^{-3}$. The $\sqrt{N} \cdot \mathrm{std}$ column reaches the neighborhood of the asymptote $\sqrt{44.31} = 6.66$ from $N = 10^4$ on (6.13, 5.88, 7.23), though at $N = 10^4$ and 3×10^4 it sits below it by more than its bootstrap standard error, and the batch-of-50 ranges are again wide: at $N = 10^5$ one batch of fifty returned 14.6. A plug-in $\hat{\chi}^2$ from a single $N = 2 \times 10^5$ draw reads 114 against the exact 162.

9.6 Grid and convention dependence of the estimand

The grid is the model. The estimand $F^*(\nu) = \mathbb{E}_{Q_{\nu,M}}[F]$ depends on the grid size M, on the temperature λ, on the proposal, and on the discretization convention of (7), and nothing in the paper establishes invariance under any of them. Under refinement of M at fixed λ the selected measures converge—in the Gaussian example with $O(\delta t)$ discretization error toward an explicit

N	|bias|	std($\hat{F}_N$)	$\sqrt{N} \cdot$ std ($\pm$ s.e.)	batch-of-50 range	median $\hat{N}_{\text{eff}}$
100	1.42×10^{-2}	2.68×10^{-1}	2.68 ± 0.11	[1.63, 3.58]	4.9
300	6.94×10^{-3}	2.07×10^{-1}	3.59 ± 0.14	[2.67, 5.36]	8.5
1 000	3.53×10^{-3}	1.37×10^{-1}	4.33 ± 0.14	[3.64, 5.32]	18.0
3 000	4.91×10^{-3}	9.18×10^{-2}	5.03 ± 0.23	[3.84, 7.10]	40.6
10 000	1.03×10^{-3}	6.13×10^{-2}	6.13 ± 0.24	[4.23, 8.07]	99.6
30 000	2.66×10^{-3}	3.39×10^{-2}	5.88 ± 0.16	[4.18, 7.11]	253.0
100 000	5.09×10^{-4}	2.29×10^{-2}	7.23 ± 0.61	[4.90, 14.56]	732.9

Table 2: Example II (double-well surrogate, $\lambda = 3/4$): convergence of $\hat{F}_N(\nu)$ over 1,000 replications (seed 20260915) against the quadrature benchmark $F^*(\nu) = -0.547760$; columns as in Table 1. CLT asymptote $\sqrt{\sigma^2(\nu)} = 6.66$.

limit (§9.4)—but to a law not representable from the proposal; the family $\{Q_{\nu,M}\}_M$ is not projectively consistent, and no telescoping identity across nested grids is available. Under the midpoint (Stratonovich-type) convention in place of the left-point one, the kinetic sum changes at order δt per step and the selected measure with it. Under a change of coordinates the classical action for state-dependent diffusion $\sigma(x)$ acquires convention-dependent terms, and the restriction to constant σ_ν keeps (7) free of such terms; the state-dependent case is out of scope. So M, λ, the proposal and the convention are part of the definition of the quantity being estimated, not numerical settings, and any empirical use of the estimator must report sensitivity to all four alongside the estimate. Example I gives the sensitivity to M and λ explicitly: the displacement of the estimand from the proposal's value is 0.280 at $(M, \lambda) = (50, 1)$ and, along the temperature schedules of §11 that hold the chi-square divergence fixed under refinement, decays toward zero.

9.7 Contents of the examples

Both examples use textbook quantities: a discretized one-dimensional OU proposal released from the origin, an OU or double-well surrogate, a quadratic terminal observable, and a forward Euler–Maruyama discretizer. No details of any production algorithm are present. Every quantity reported in §9.2–§9.5 is regenerated by the archived script described in the Declarations.

10 Structural comparison with classical analogues

We compare the selection-functional construction with four frameworks: Girsanov path reweighting, misspecified-model theory, path-space variational inference, and generalized Bayes. The comparison is summarized in Table 3. Two statements organize it. The first is the corrected exclusion (E1) of Remark 2.2: no Markov generator on the observed state space reproduces the target's two-time family. The second is Proposition 3.5: the selection weight is a single tempered path density, not a ratio. Neither is a statement about what exists; both are statements about what the construction does.

10.1 Girsanov reweighting of externally generated path ensembles

The closest prior art is the reweighting of externally generated path ensembles by a path action, which is established practice, rather than any optimization-based method. Donati, Hartmann and Keller [24] generate a path ensemble under a reference potential and reweight it to obtain

path-ensemble averages—and Markov state models—under a perturbed potential, using the Girsanov change of measure on path space; the paths are produced by a separate simulation and then reweighted, which is structurally the operation of Definition 3.10. Adib [1] shows that the two standard discretized stochastic actions, with and without the second-derivative term corresponding to the divergence sum of (7), are equivalent for the purposes of reweighting and sampling trajectories, and carries out the reweighting of an ensemble generated under one dynamics to another. Both use a *difference* of two actions, equivalently a ratio of two path densities with a common diffusion coefficient; their weights are likelihood ratios, so they remain bounded in chi-square under refinement (§9.4, third row of the refinement table) and the divergence term cancels or is immaterial for them. The present construction uses a *single* action and forms a product of densities. Products of densities are themselves established (the product of experts of [39]), so the novelty claim of this paper does not rest on multiplying rather than dividing densities; it rests on the exact product identity (Proposition 3.5), its finite-grid implementation, and the consequences derived from it here: Corollary 3.6, the three consequences of §3.4, the refinement mechanism of §9.4 and the design rule of §7.5. The "third operational role" framing of the July version, which did not position the construction against this literature, is withdrawn. Whether a product is the right object is a modeling question, answered in §11: a product is the right object when the proposal and the surrogate are each partially informative and neither is a reference measure for the other.

10.2 Misspecified-model theory

White's analysis of the quasi-maximum-likelihood estimator under misspecification [79] establishes that, under regularity conditions, $\hat{\theta}_n$ converges to the pseudo-true value $\theta_* = \arg\min_\theta D_{\mathrm{KL}}(P_0 \,\|\, P_\theta)$, where P_0 is the true law and $\{P_\theta\}$ the fitted family, with sandwich-covariance asymptotic normality [43, 58]; the Bayesian analogue [46, 47, 77, 35] establishes Bernstein–von Mises theorems under misspecification. The structural assumption is that a true law P_0 exists against which the family is scored. The July version claimed a distinction here—that under indivisibility "there is no generator-based P_0 to score against"—and the claim was wrong: misspecified-model theory does not require P_0 to belong to any generator-based family; its purpose is to study estimation when P_0 lies outside the fitted family, and setting (L) supplies $P_{\mathrm{true},M}$ as P_0. The calibration of §8.4 is, accordingly, an instance of misspecified-model theory and is presented as one. The distinction is that the *selection* step is not a fit. The surrogate is not optimized toward the target at all; it is held fixed as one factor of a product measure, and Proposition 7.2 says that optimizing its kernels toward an indivisible target would in any case hit a deficit-determined floor.

10.3 Path-space variational inference

Variational inference [76, 8] targets an intractable but well-defined posterior by minimizing a divergence over a tractable family; importance-weighted variants [11, 54, 59, 23] tighten the bound with multi-sample estimators, and chi-square [21] and Rényi [50] divergences extend the divergence family. Path-space instances include Schrödinger-bridge formulations [49, 18, 62], control-based importance sampling of diffusions [37, 80], stochastic-optimal-control matching [22], and the path-space variational coarse-graining of [36], which projects a non-Markovian molecular dynamics onto a Markovian surrogate by minimizing a path-space relative entropy. The July version claimed that such a relative entropy "presupposes a generator-derived density evaluable along paths" and that under indivisibility there is "nothing for a variational method to approximate." Neither is right: relative entropy between path measures requires probability measures and absolute continuity, not generators, and non-Markovian path laws have Radon–Nikodym derivatives with respect to suit-

able reference measures. The true statement is narrower. These are *optimization* constructions: they move the surrogate toward the target in a divergence, and they need the target's path law (or samples from it) to do so. The selection-functional construction is not a variational projection; it holds the surrogate fixed, needs no target law, and its target is the selected measure itself.

10.4 Generalized Bayes

The Bissiri–Holmes–Walker framework [7] characterizes the coherent update of a prior by a loss ℓ as $p^{(\mathrm{GB})} \propto e^{-\lambda R}\,\Pi$, with R the cumulative loss evaluated at observed data. The selected measure (12) has this algebraic form with $\ell = S_{\nu,M}$, and the resemblance ends there (Remark 3.11): the candidate path is not data conditioning a parameter, it is the variable being distributed, and no observation enters. The July version further asserted that PAC-Bayes risk bounds for Gibbs posteriors [13, 3] "transfer to the calibrated surrogate level"; they do not, because the calibrated object $\hat{\nu}_{\mathrm{QML}}$ is a point estimator, not a Gibbs posterior over ν, and no prior on Θ, no posterior over ν and no empirical-risk setting is formulated in this paper. That assertion is withdrawn.

10.5 Summary table

	Girsanov path reweighting	Misspecified-model theory	Variational inference	Selection functional (this paper)
Setting	(L): reference and target path laws	(L): true law P_0	(L): posterior on path space	Theorems A–C: neither (only $Q_{\nu,M}$); Theorem D: (L); Prop. 7.2: (K)
Weight / objective	ratio $p_{\mathrm{target}}/p_{\mathrm{ref}}$; difference of actions	KL projection θ_*	KL-closest q^*_ϕ in a tractable family	single action: product $\Pi_M \cdot p_\nu^\lambda$ (Prop. 3.5)
Quadratic variation	preserved (common σ)	—	—	changed: $1/\sigma^2_{\mathrm{eff}} = 1/\sigma^2_\Pi + \lambda/\sigma^2_\nu$
χ^2 under refinement	bounded	—	—	exponential in M
Fitting target	target path law	θ_*	q^*_ϕ	$Q_{\nu,M}$; estimand $F^*(\nu)$; surrogate held fixed
Truth link	definitional	definitional (P_0 assumed)	definitional (posterior assumed)	conditional transfer (D1); design rule (23); kernel obstruction (22) independent of it

Table 3: Structural comparison. The column on the right differs from the others in two ways, neither of which is a claim that a true law, a likelihood or a posterior fails to exist: (a) no Markov generator on the observed state space reproduces the target family, so no generator-based path density is available on that description (exclusion (E1)); (b) the weight is a single tempered path density rather than a ratio, so the selected measure's quadratic variation is set by the pair and not inherited from a reference. The July version's table listed two "structural distinctions" that do not survive (no true law to score against; no likelihood for a variational method to approximate) and is replaced.

11 Discussion and open problems

An externally proposed candidate-path ensemble on a finite grid is reweighted by a single tempered surrogate path density with an Onsager–Machlup divergence correction, and the self-normalized estimator of expectations under the resulting product measure is consistent, asymptotically normal, and concentrates at the rates the two weight regimes allow (Theorems A–C). The identity behind the construction (Proposition 3.5) fixes the selected measure's diffusion coefficient, and with it the paper's degeneracies, its finite-grid character, and the condition under which the transfer to the true process can hold at all (Theorem D, Proposition 7.5); the kernel-level obstruction (Proposition 7.2) is a separate fact, and the paper no longer connects them. Two examples evaluate every object against exact benchmarks, one of them exercising the divergence term that makes the action Onsager–Machlup rather than Euler.

The use of the estimand The estimator returns $F^*(\nu)$, the expectation under a product of experts of the proposal law and the tempered surrogate law. When (D1) can be established, Theorem D transfers it. When it cannot—which, for a proposal with the target's own roughness at a fine grid, is the typical case by Proposition 7.5—$F^*(\nu)$ is a consensus readout of two partially informative objects, a candidate-path generator (the proposal) and a surrogate law, each of which would be wrong on its own and neither of which is a reference measure for the other. It is consumed as a point readout where such a consensus is what the application wants, at a coarse grid and a temperature chosen with the diagnostics of Theorem C in view, and with the sensitivity to M, λ, the proposal and the convention reported alongside it (§9.6). Its decision-theoretic value is application-specific and this paper does not assert one.

Four problems are left open, and one of the July version's five is closed. The first is restated in the form the present paper shows it must take; the second records the closure.

The truth-link bridge, as a design problem For the admissible class of proposal–surrogate–temperature triples whose selected quadratic variation matches the target's (Proposition 7.5), control the residual tolerance $\eta_{\nu,M}$ of (D1) at fixed M by the target's indivisibility invariants, such as the integrated deficit $J(\mathcal{R})$ of [70, Theorem 7.4], together with separately measured proposal and surrogate errors (Remark 7.6). Stated without the class, it is not well posed.

The continuum selected measure exists; its fixed-proposal importance-sampling representation does not The July version deferred a "renormalized continuous-path theory" to future work and proposed a temperature schedule $\lambda_M \downarrow 0$ calibrated to the refinement blow-up. Two things are now clear. First, at fixed λ the selected measures $Q_{\nu,M}$ converge: in Example I the limit is an explicit Ornstein–Uhlenbeck law with the diffusion coefficient of Corollary 3.6, and the estimand converges to it with $O(\delta t)$ error (§9.4). The fixed-proposal estimator has no non-degenerate limit: the limit law is singular to the proposal's, so the importance weights from Π_M degenerate, and the refinement table measures that degeneration. This does not exclude a joint limit in which N grows with M at least as fast as $1 + \chi^2(Q_{\nu,M} \,\|\, \Pi_M)$, or a sequence of proposals adapted to the selected law; neither is studied here. Second, the temperature-schedule route does not survive testing. In Example I, solving for the schedule that holds the chi-square divergence at a fixed value and measuring the selection that remains—the displacement $|F^*(\lambda_M) - F^*_\Pi|$ of the estimand from its proposal value—gives:

M	50	100	200	400	800
λ_M ($\chi^2 \equiv 10$)	0.4307	0.2750	0.1815	0.1224	0.0837
displacement	0.1903	0.1438	0.1069	0.0784	0.0569
λ_M ($\chi^2 \equiv 1372.4$)	1.0000	0.5777	0.3581	0.2318	0.1544
displacement	0.2800	0.2210	0.1697	0.1275	0.0943

In both schedules the displacement decays toward zero: by Corollary 3.6 the selection strength and the singularity are controlled by the same parameter, so a schedule that keeps the estimator usable drives the construction toward the proposal itself. Since $\sigma_{\text{eff}} < \sigma_\Pi$ for every $\lambda > 0$, the selected measure is singular to the proposal in the continuum for every fixed temperature. The natural renormalization that restores absolute continuity—subtracting the proposal's kinetic term from the action—converts the weight into a Girsanov ratio and the construction into the established one of §10.1; we know of no other. The single-action *estimator* is therefore intrinsically a finite-grid object: its continuum limit is either trivial or standard, even though the measure it targets has a limit. This is stated as a conclusion, not an open problem, and the finite grid is the model for that reason.

Variance-optimal surrogate for an observable class Characterize $\nu^\dagger$ for a class $\mathcal{F}$ under a stated objective ($\sup_{F\in\mathcal{F}} \sigma_F^2(\nu)$, or an average), an unconditional variance-only optimization for which Theorem C supplies the governing quantities. Under (D1) with a known tolerance, the asymptotic trade-off between selection bias and estimator variance is $4F_{\max}^2\eta_{\nu,M}^2 + \sigma^2(\nu)/N$; the finite-$N$ estimator carries its own bias, so this is not a risk formula. For the OU surrogate the sweep is one-dimensional in the restoring rate, with the chi-square scale bounding the useful range from above; Example I gives $k^\dagger = 12.39$ for the terminal observable.

Adaptive proposals Allow the proposal Π_n to depend on the previous $n-1$ samples (sequential importance sampling or particle filtering, exploiting Remark 3.12 when the proposal factorizes sequentially). Consistency does not follow automatically: it requires conditions controlling the adaptation, the dependence, the stability of the weights and the support of the adaptive proposals, and even the fixed-proposal examples exhibit severe weight degeneracy; the central limit theorem requires martingale machinery in the spirit of [19, 16]. The conditions that would need to be proved are stated; nothing is claimed.

Multilevel selection A multilevel Monte Carlo analogue [31] across a hierarchy of grids $M_0 < M_1 < \cdots$ requires estimators on successive levels to approximate a common limiting quantity with controlled level bias and coupled level differences of decaying variance. The limiting quantity exists (previous paragraph), and the level bias is $O(\delta t)$ in the Gaussian example; the coupling is missing. Each level's estimator is an importance representation from a common proposal whose chi-square divergence grows as $e^{0.144M}$, so the level-wise variances grow rather than decay, and the selected measures are not projectively consistent (§9.6), so no telescoping identity is available. Whether coupled increments can be arranged to offset that growth is the open design question; it is recorded as conditional.

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Code availability. Every number reported in this paper that is not a citation—the closed forms and identity values of §3.4 and §3.5, Example 7.4, the tables of §7.5, the calibration sweep of §8.4, the certificate constants of §8.2, everything in §9, and the schedule table of §11—is produced by a single archived script, `paperC_numerics.py` (NumPy/SciPy), deposited with this preprint: closed forms and quadrature benchmarks are deterministic, and the Monte Carlo studies use NumPy's default generator with seeds 20260914 (Example I) and 20260915 (Example II). The closed-form quantities of Example I are additionally reproducible from the precision matrices stated in the text without the script.
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